Data Mining and Neural Networks

Unsupervised Learning

Week 4.2

Muhammad Ateeq

MA3022/MA4022/MA7022

School of Computing and Mathematical Sciences
University of Leicester

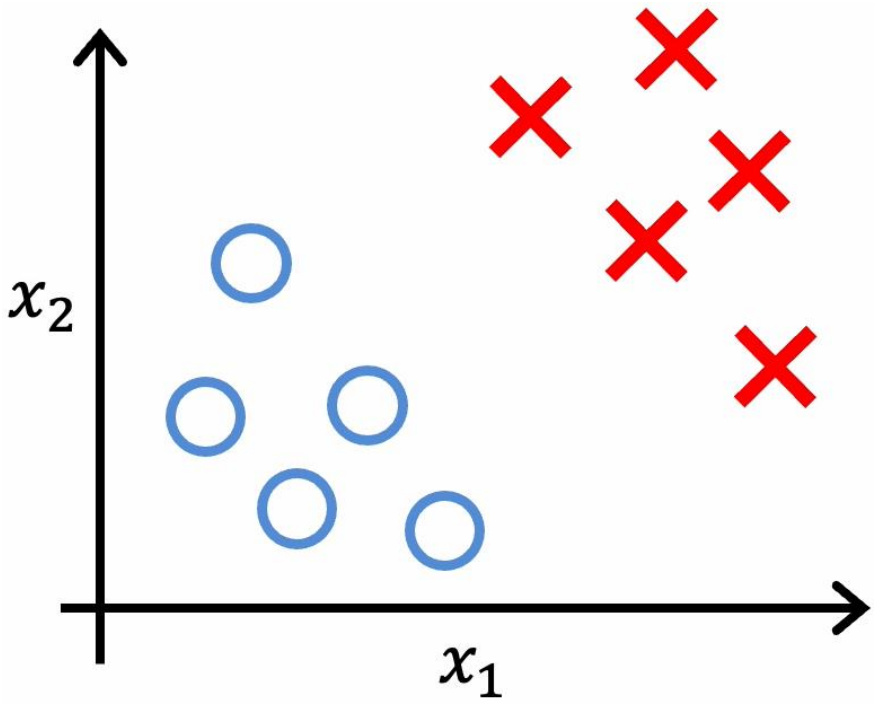
06-Feb-2026

Credits: Andrew Ng

Agenda

- **Unsupervised learning**
  - Clustering
    - Partition-based
    - Density-based
    - Hierarchical
  - Dimensionality reduction
    - Principal component analysis

Supervised Learning



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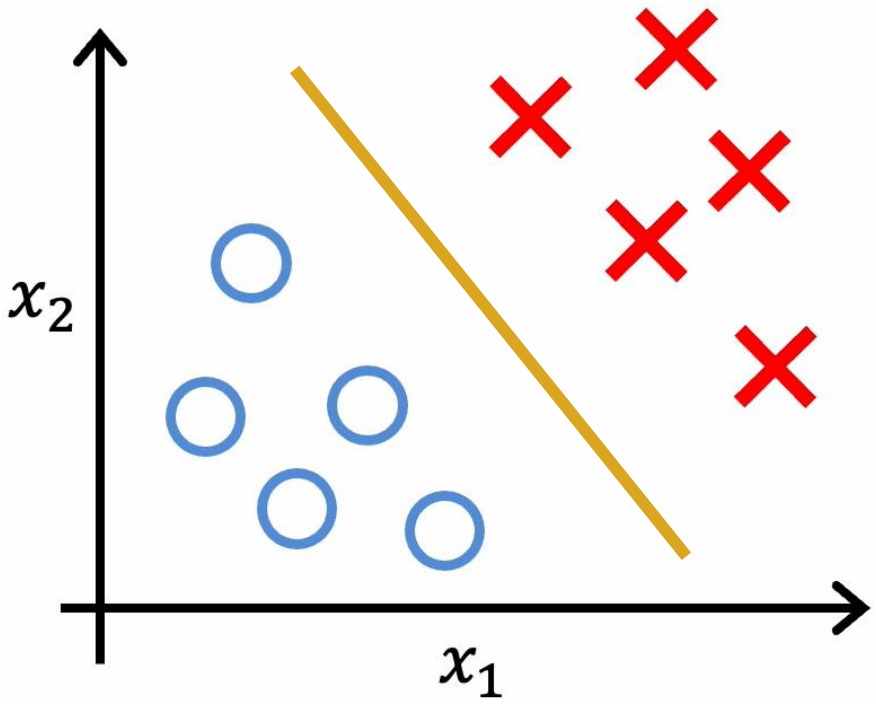

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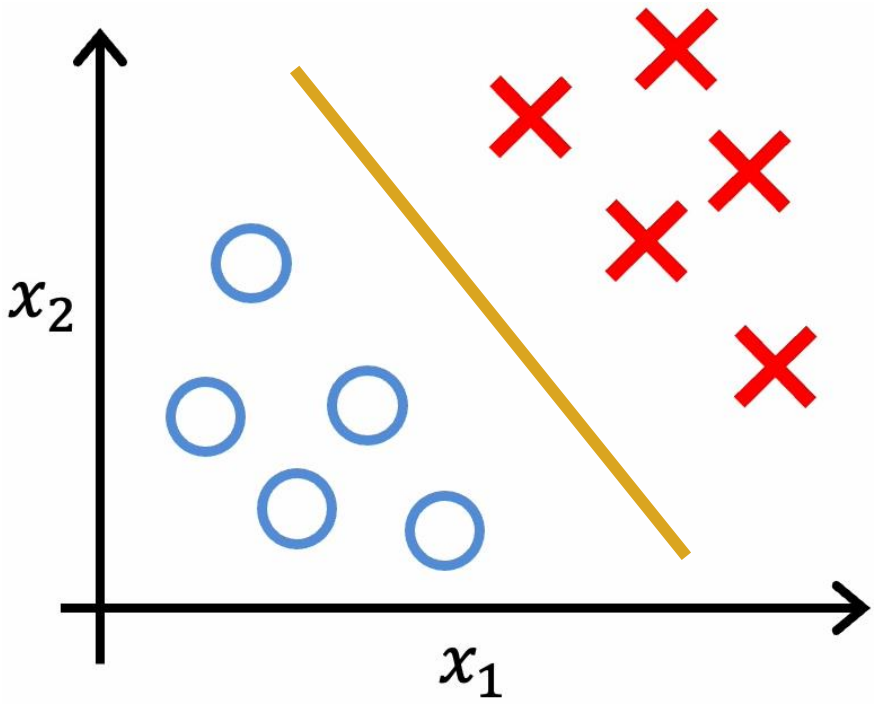

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \ldots, (x^{(m)}, y^{(m)})\}$

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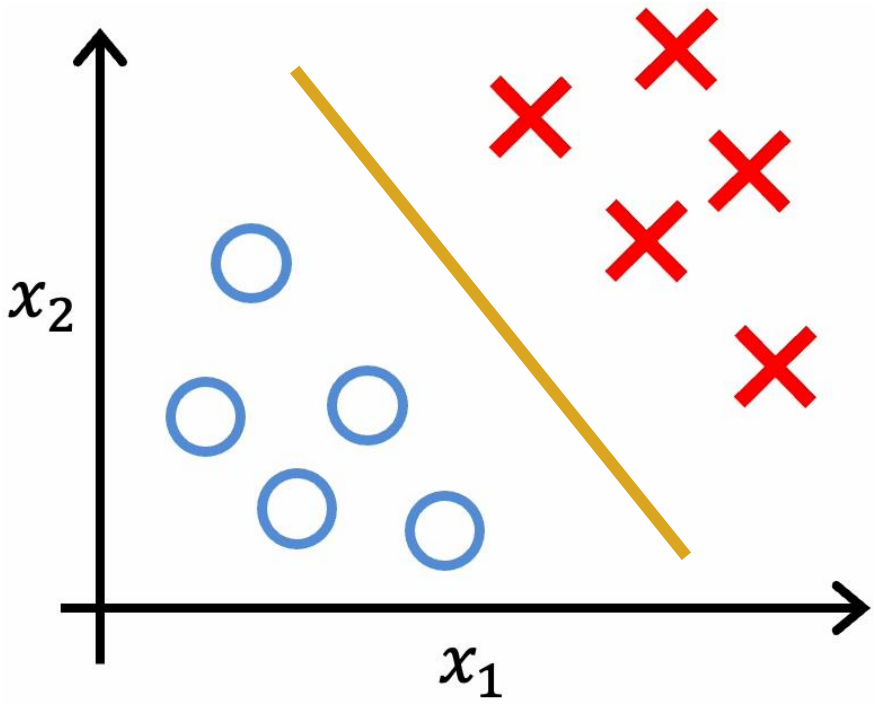

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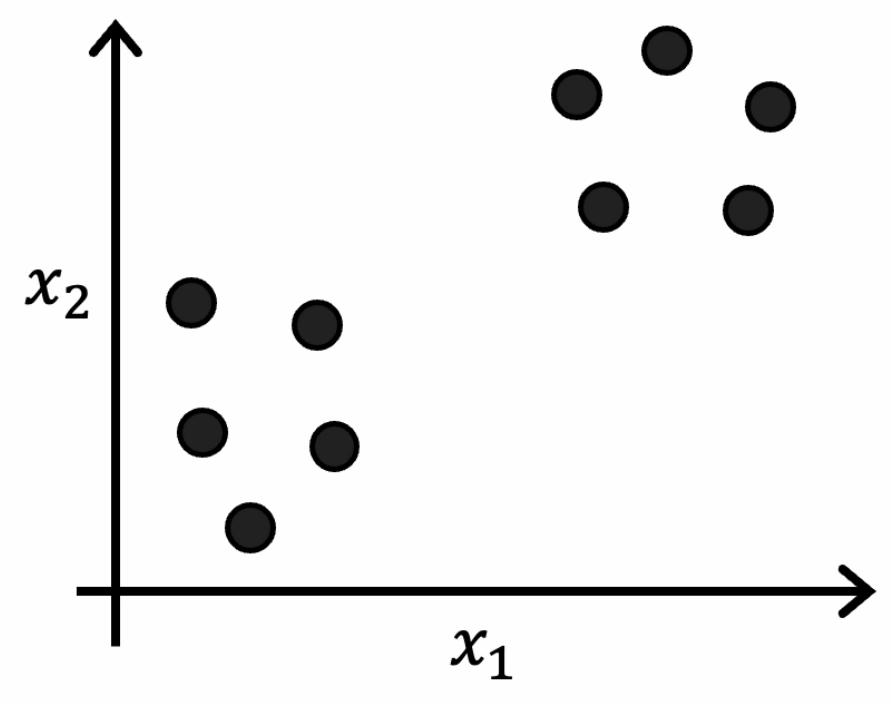

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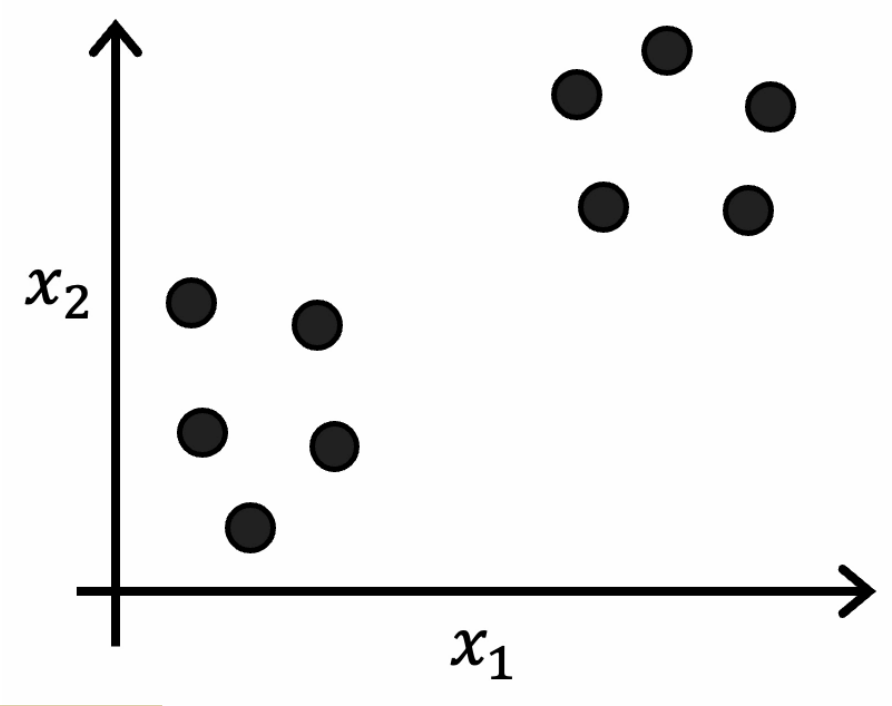

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Unsupervised Learning

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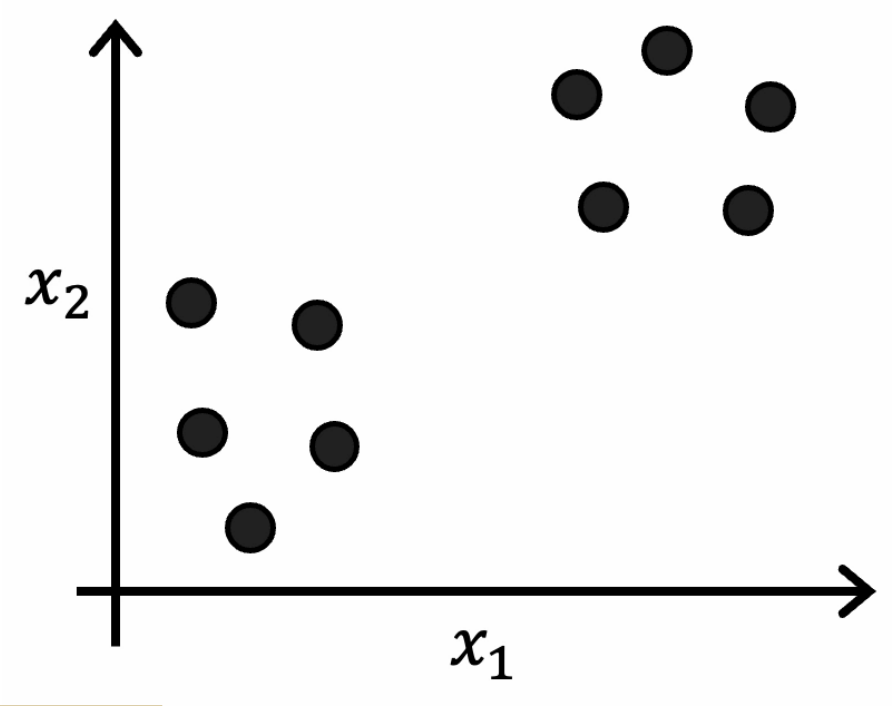

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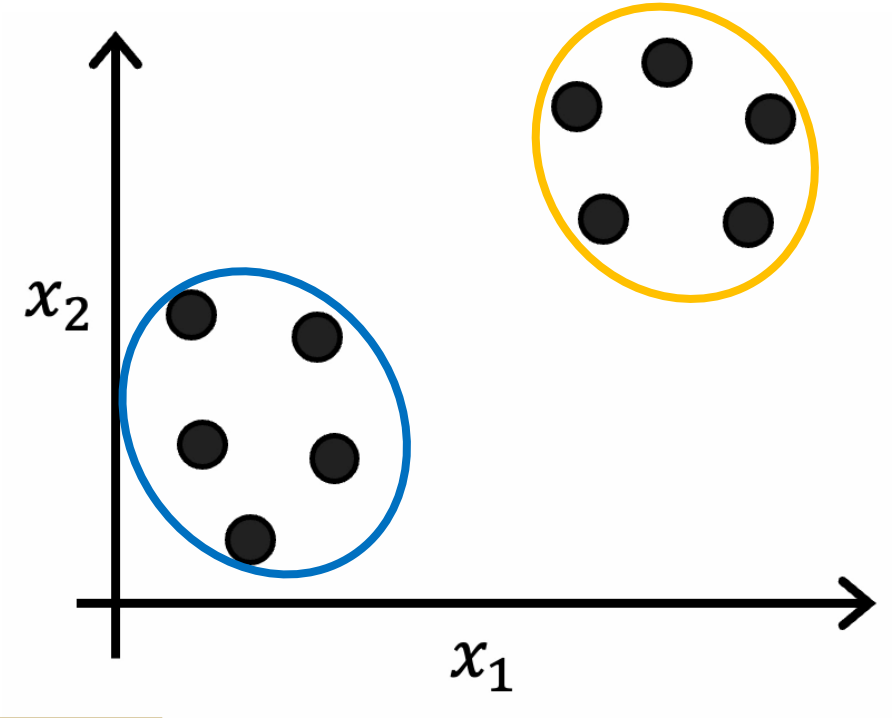

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Applications of Clustering

Grouping similar news

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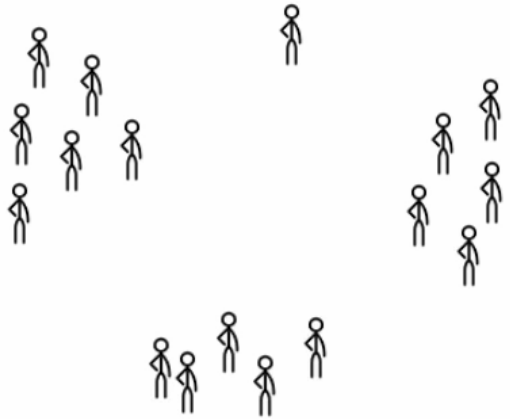

Market segmentation

Applications of Clustering

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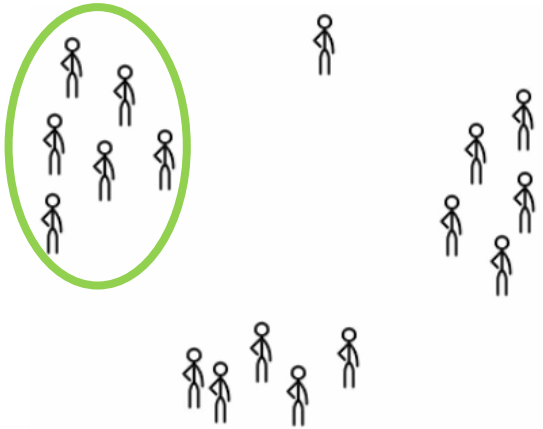
Growing skills



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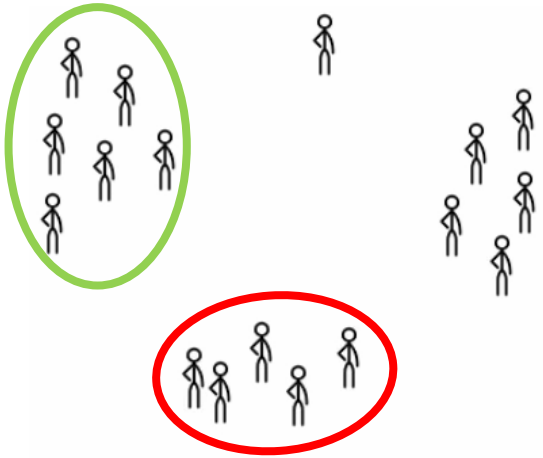
Develop career



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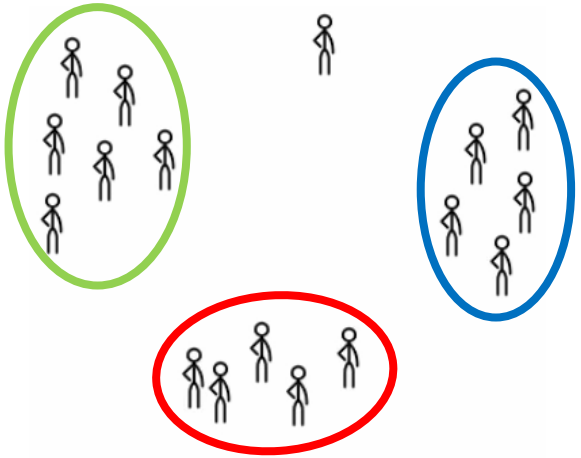
Stay updated with AI, understand how it affects your field of work



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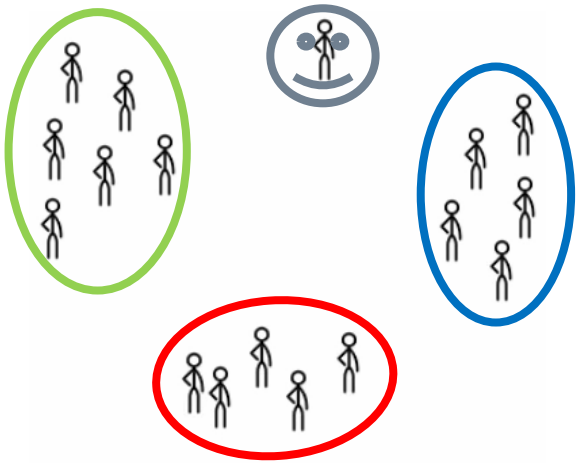

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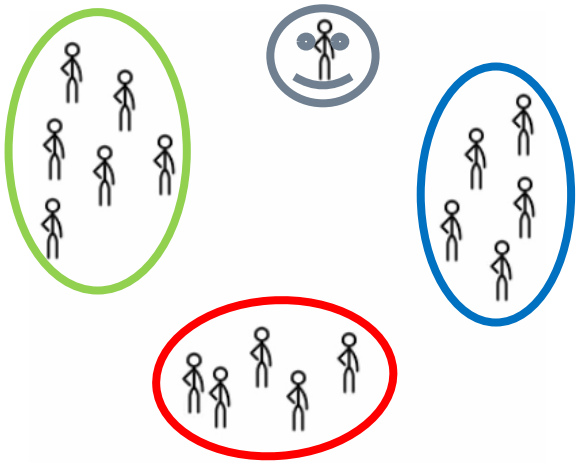

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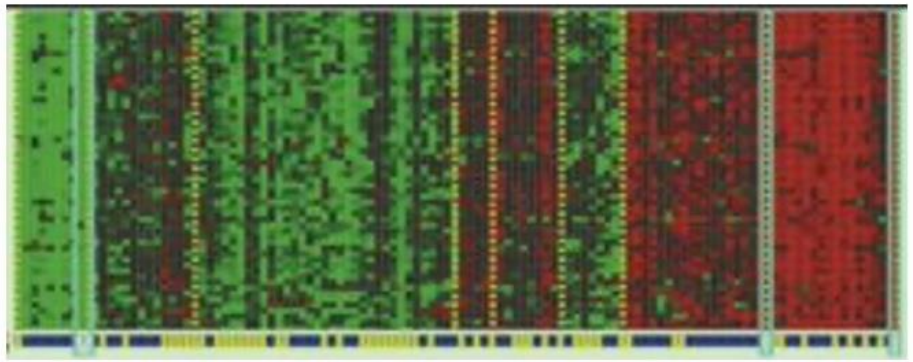

DNA analysis

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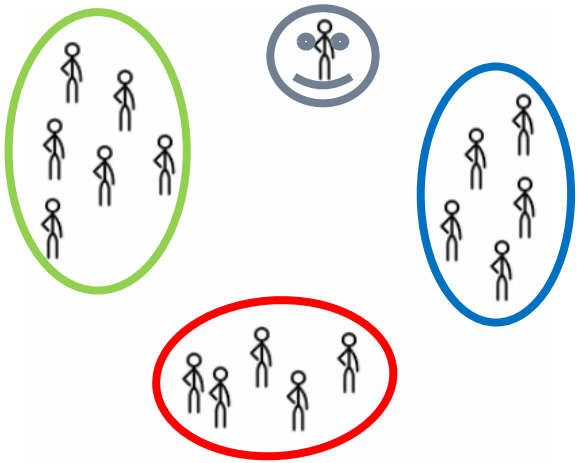

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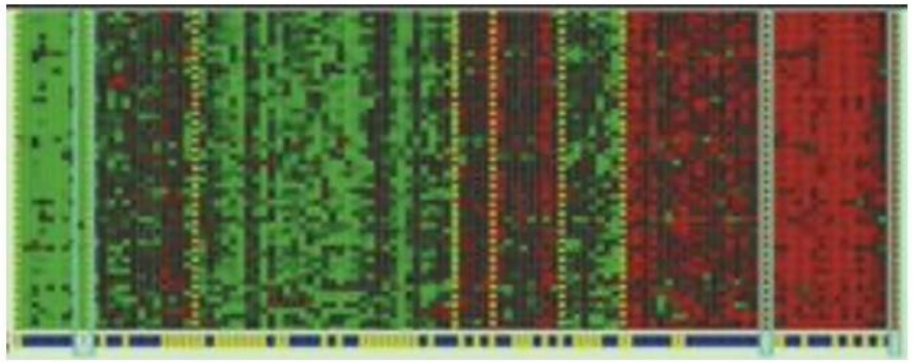

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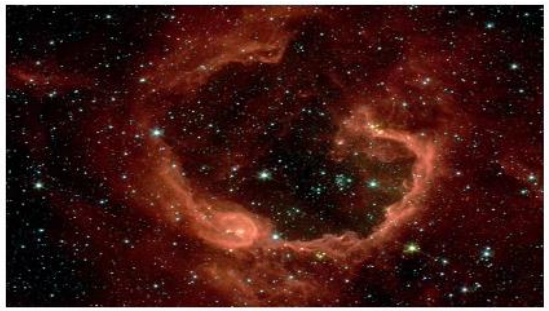

Image credit: NASA/JPL-Calte
ch/E. Churchwell (Univ. of Wisconsin, Madison)

Astronomical data analysis

Agenda

- Unsupervised learning
  - **Clustering**
    - **Partition-based: *K-Means***
    - Density-based
    - Hierarchical
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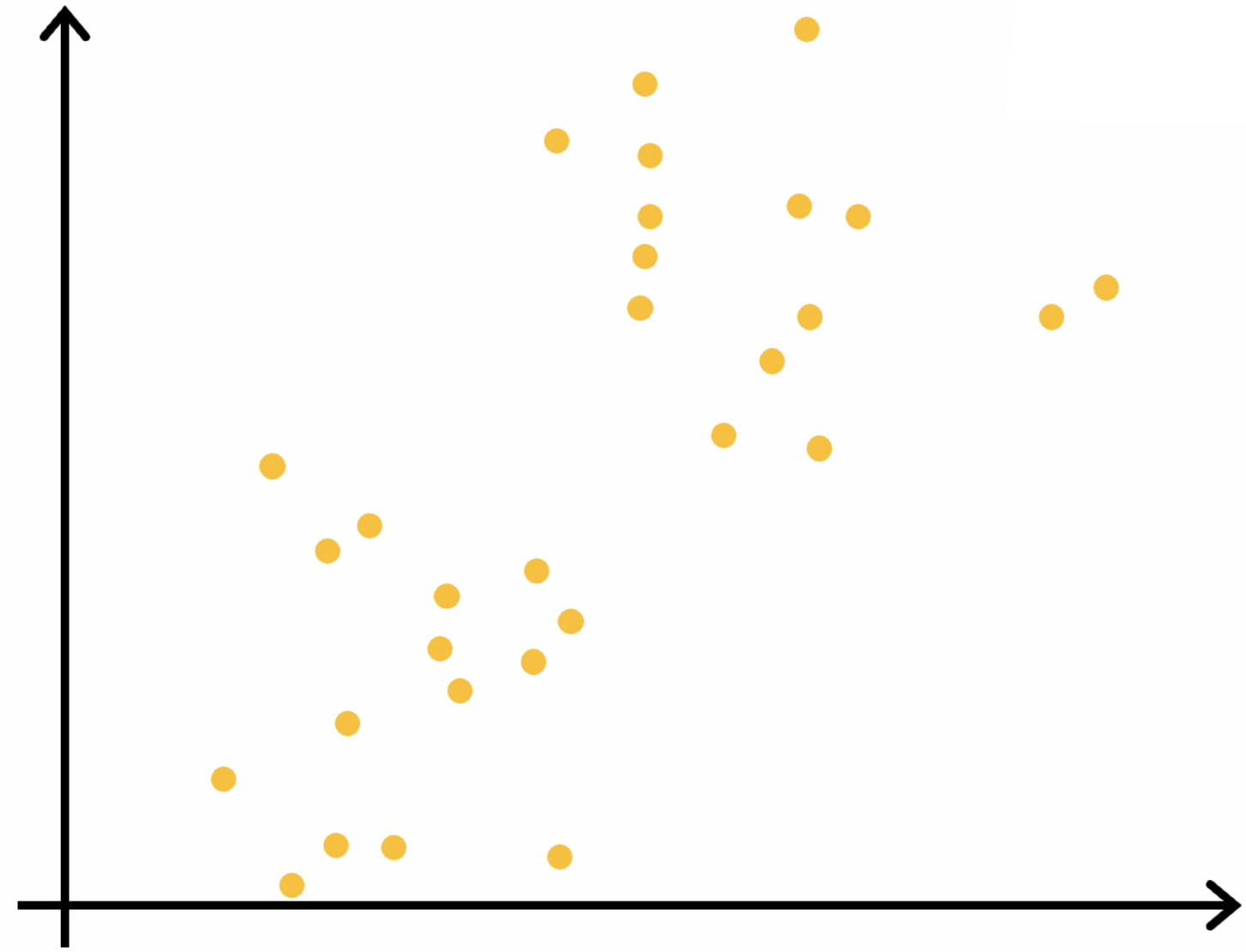

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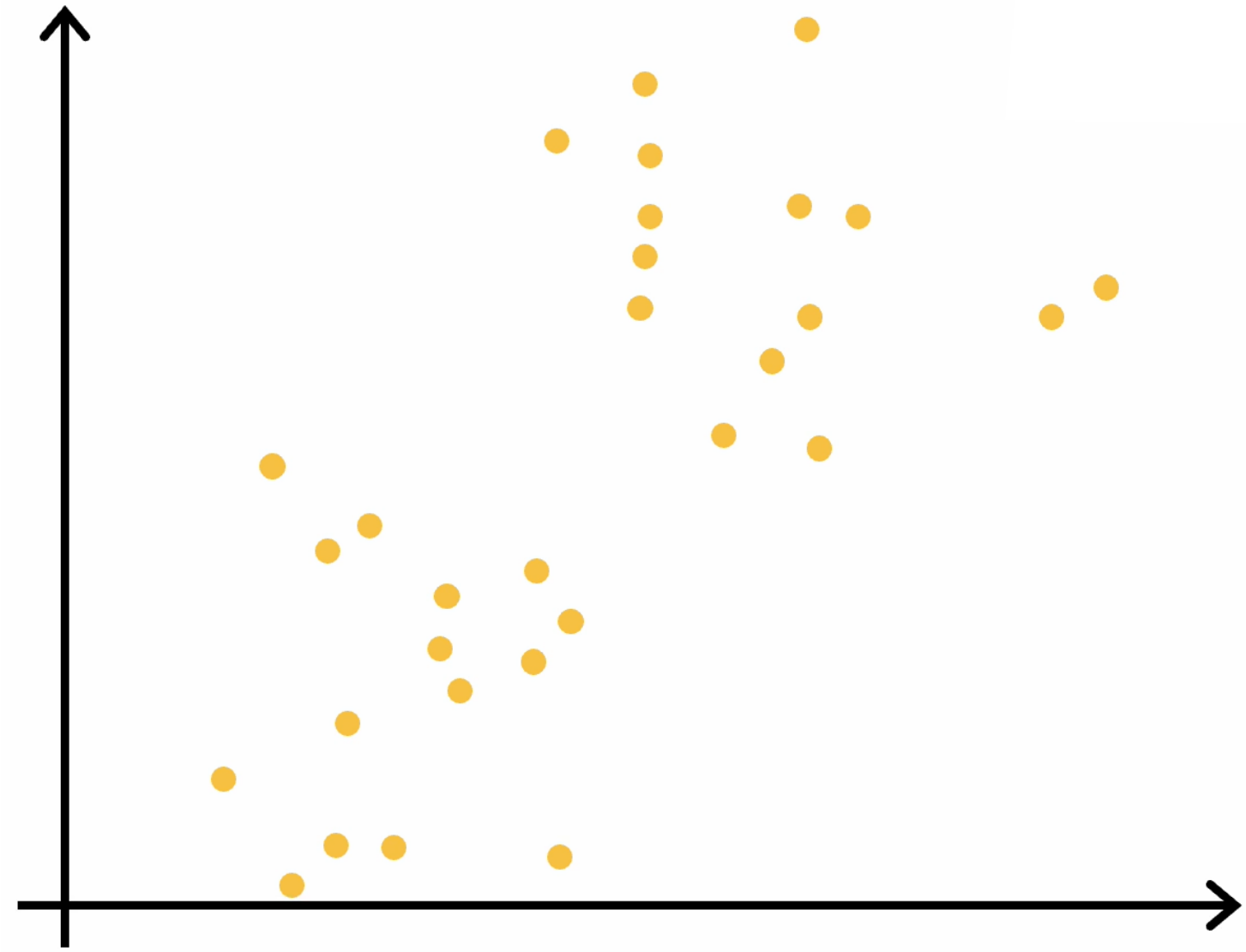

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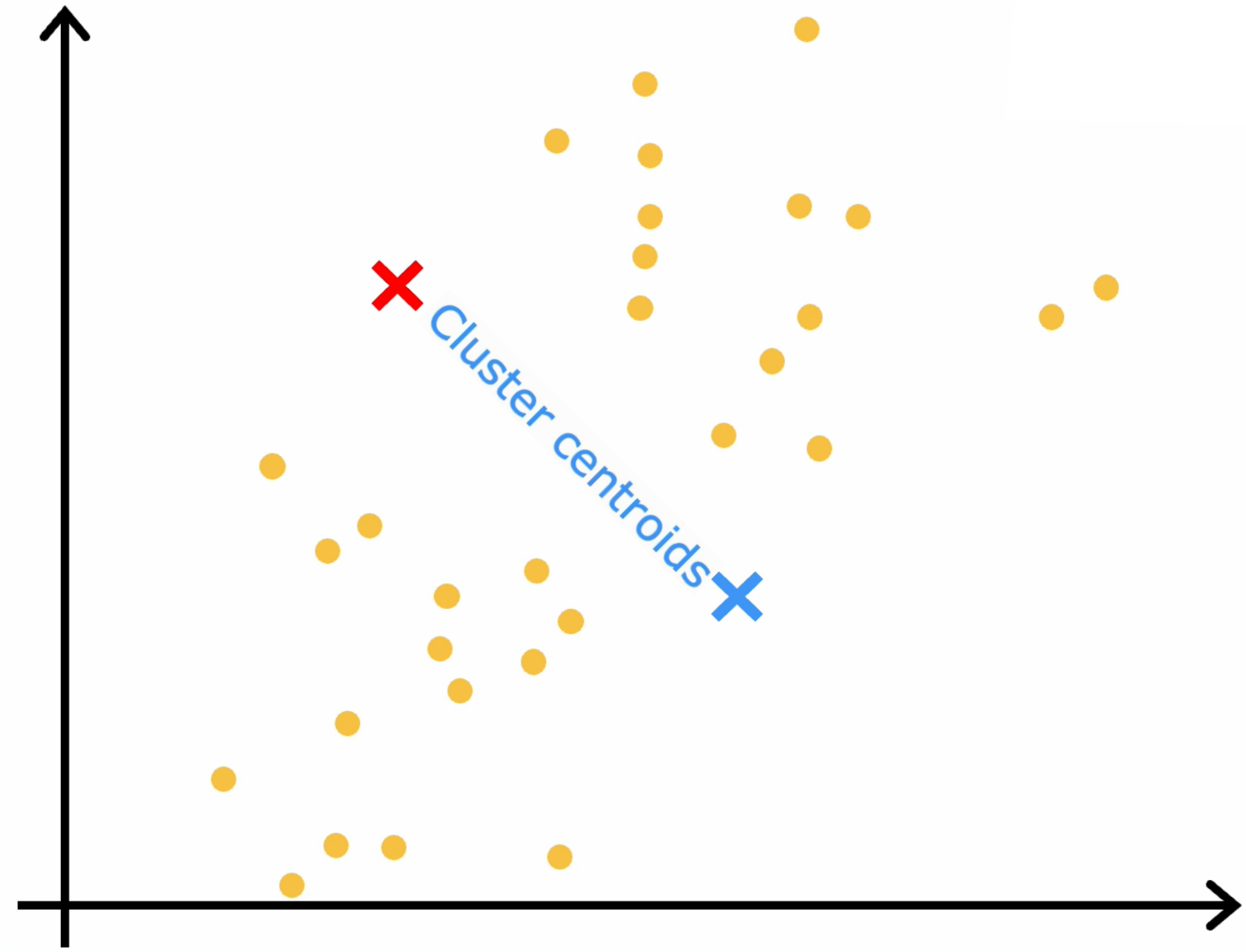

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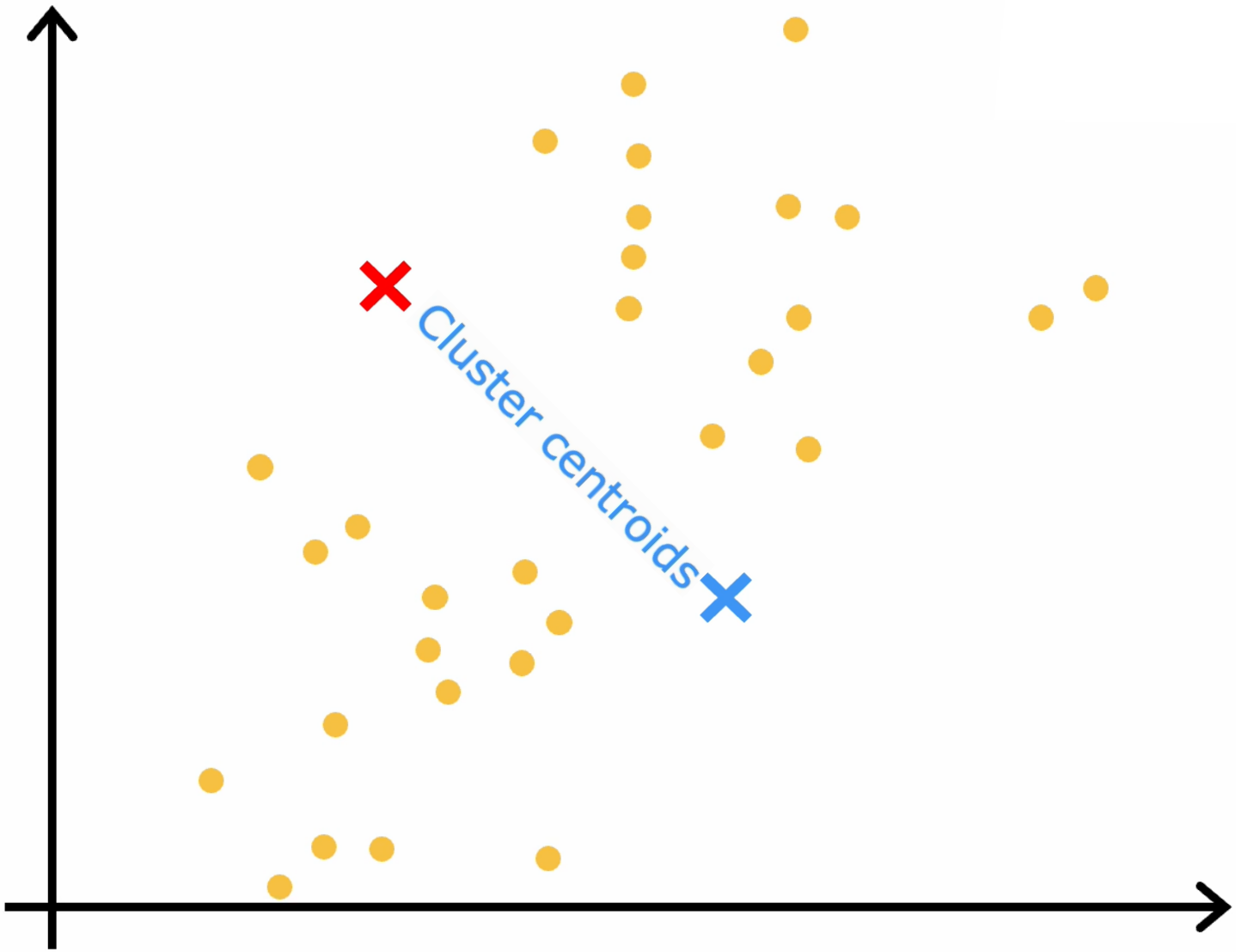
Step 1:
Assign each point to its closest centroid



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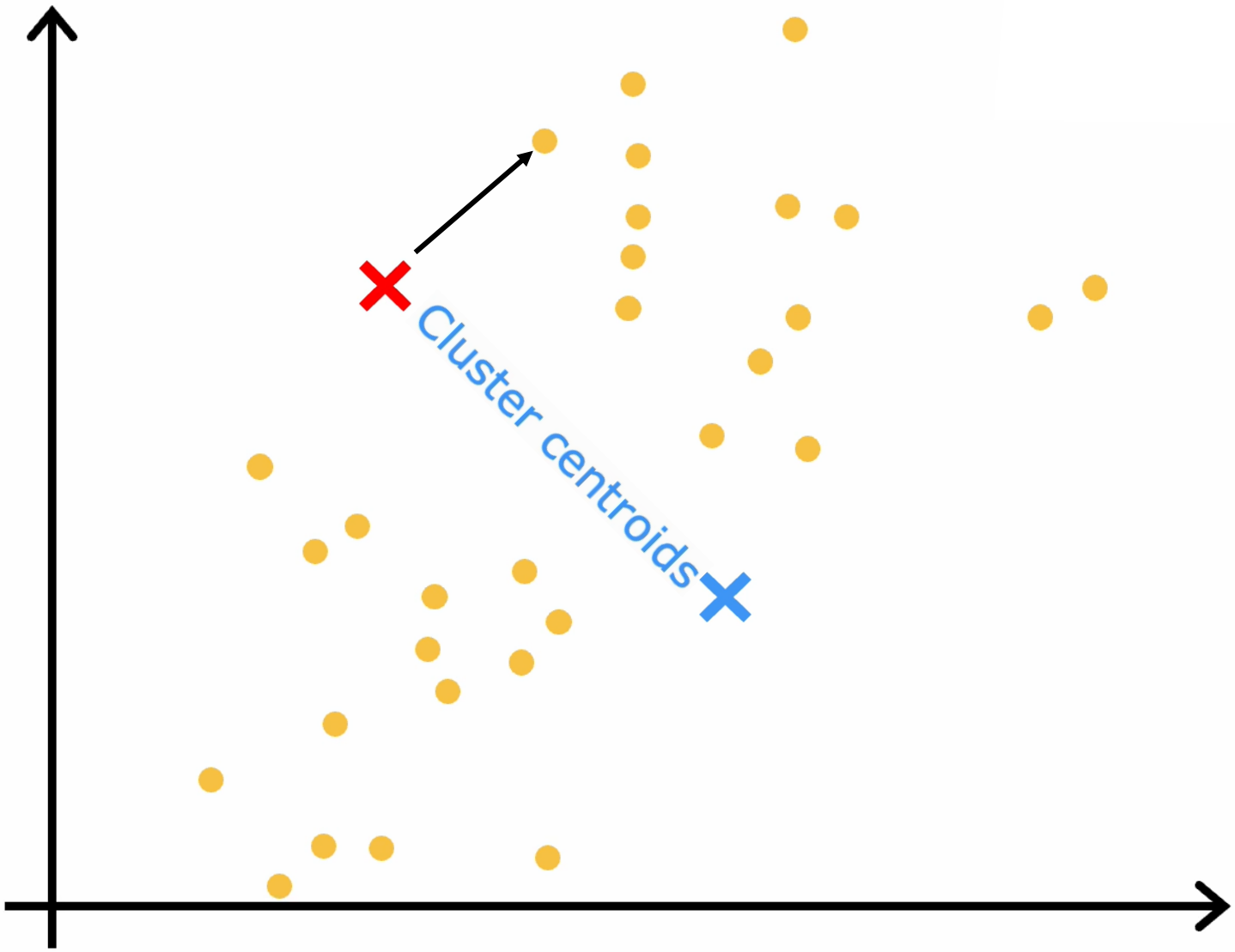

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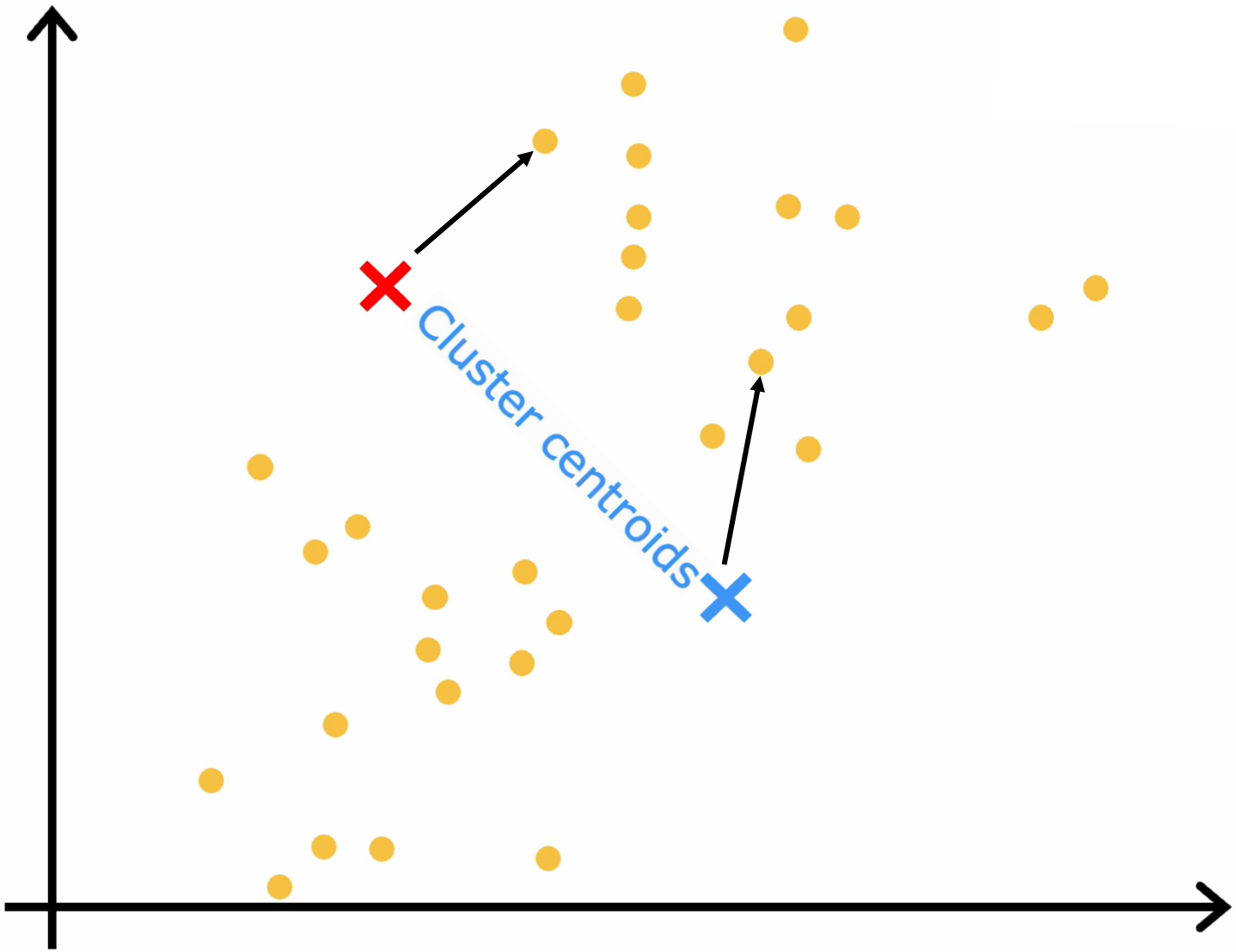

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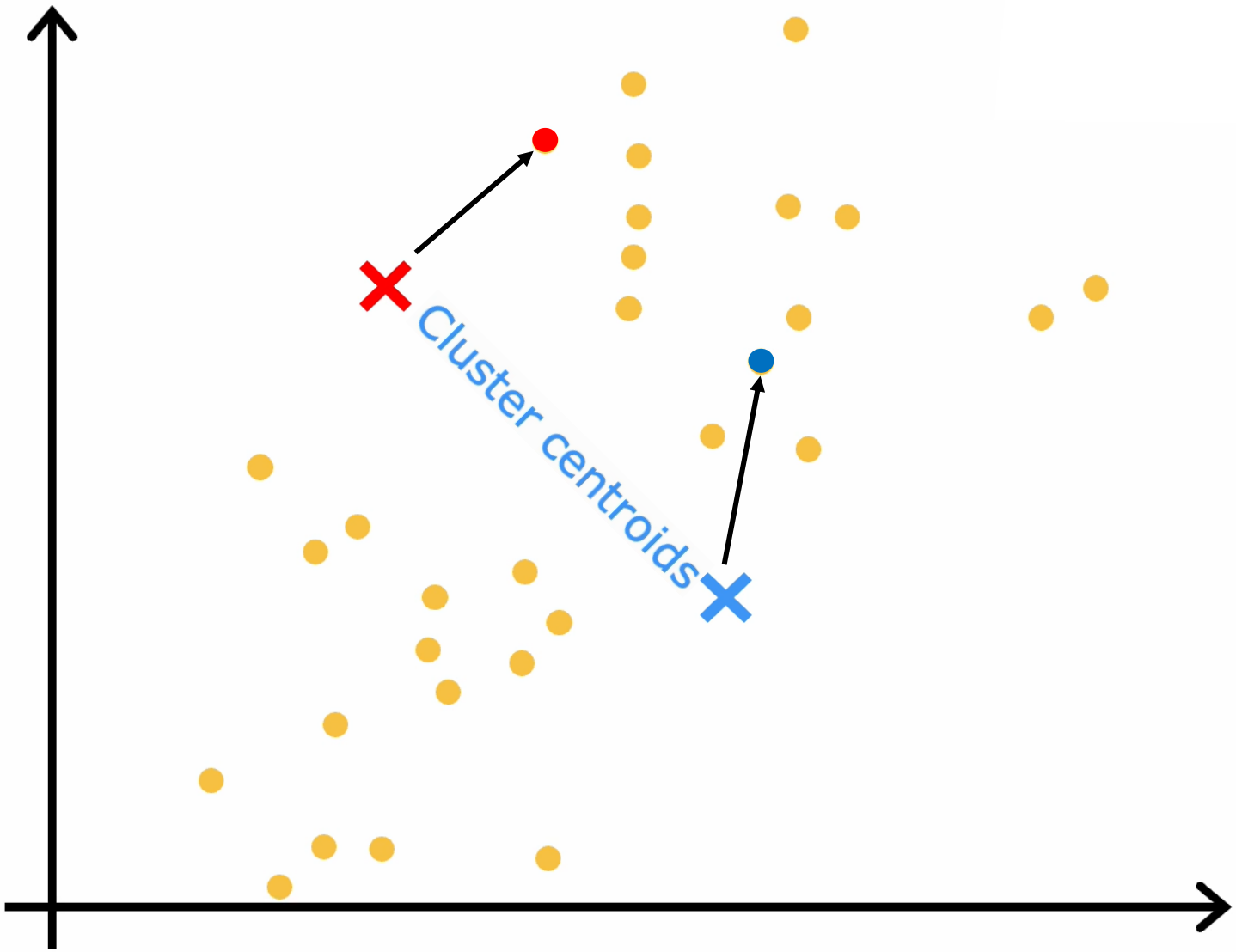

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Step 2:
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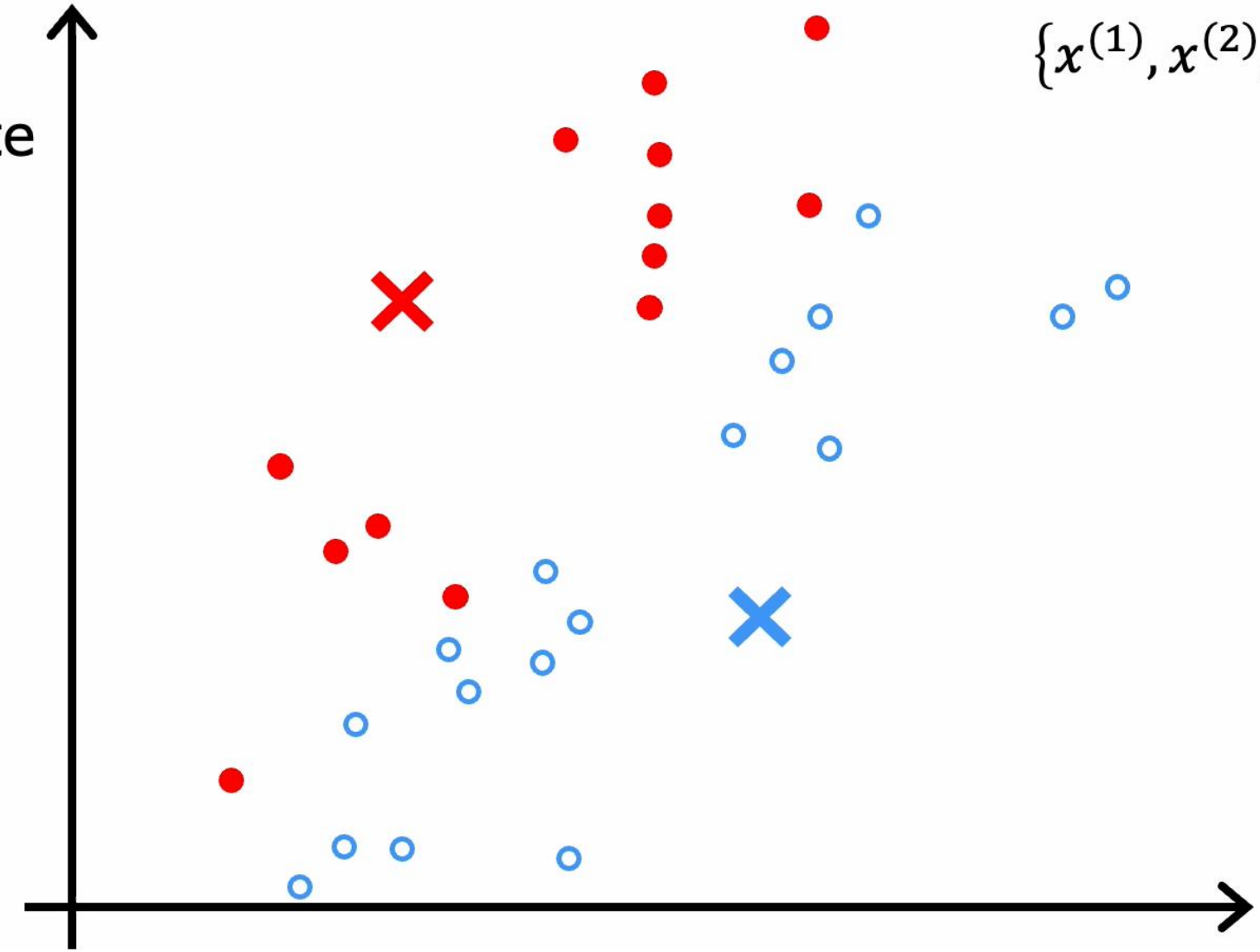

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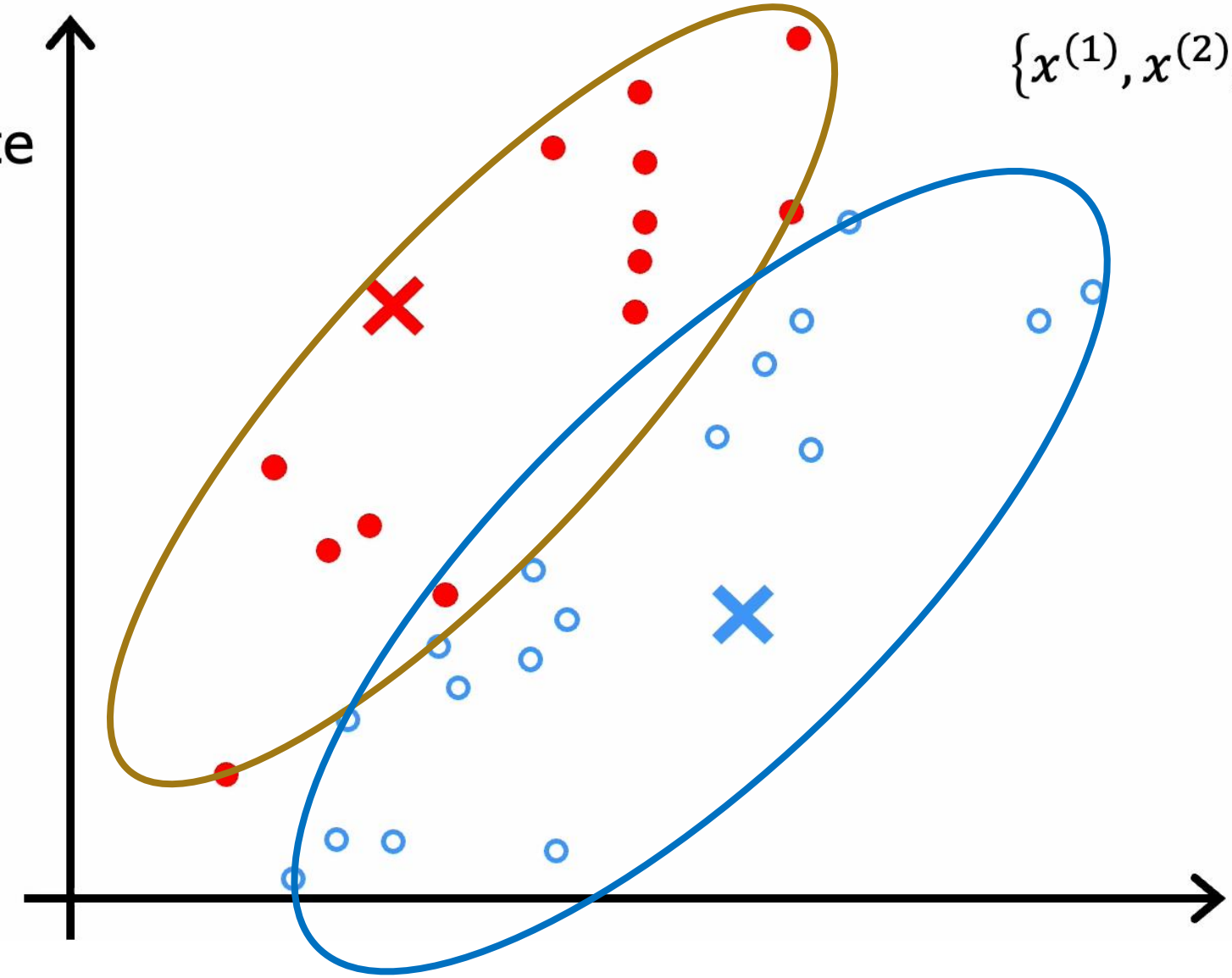

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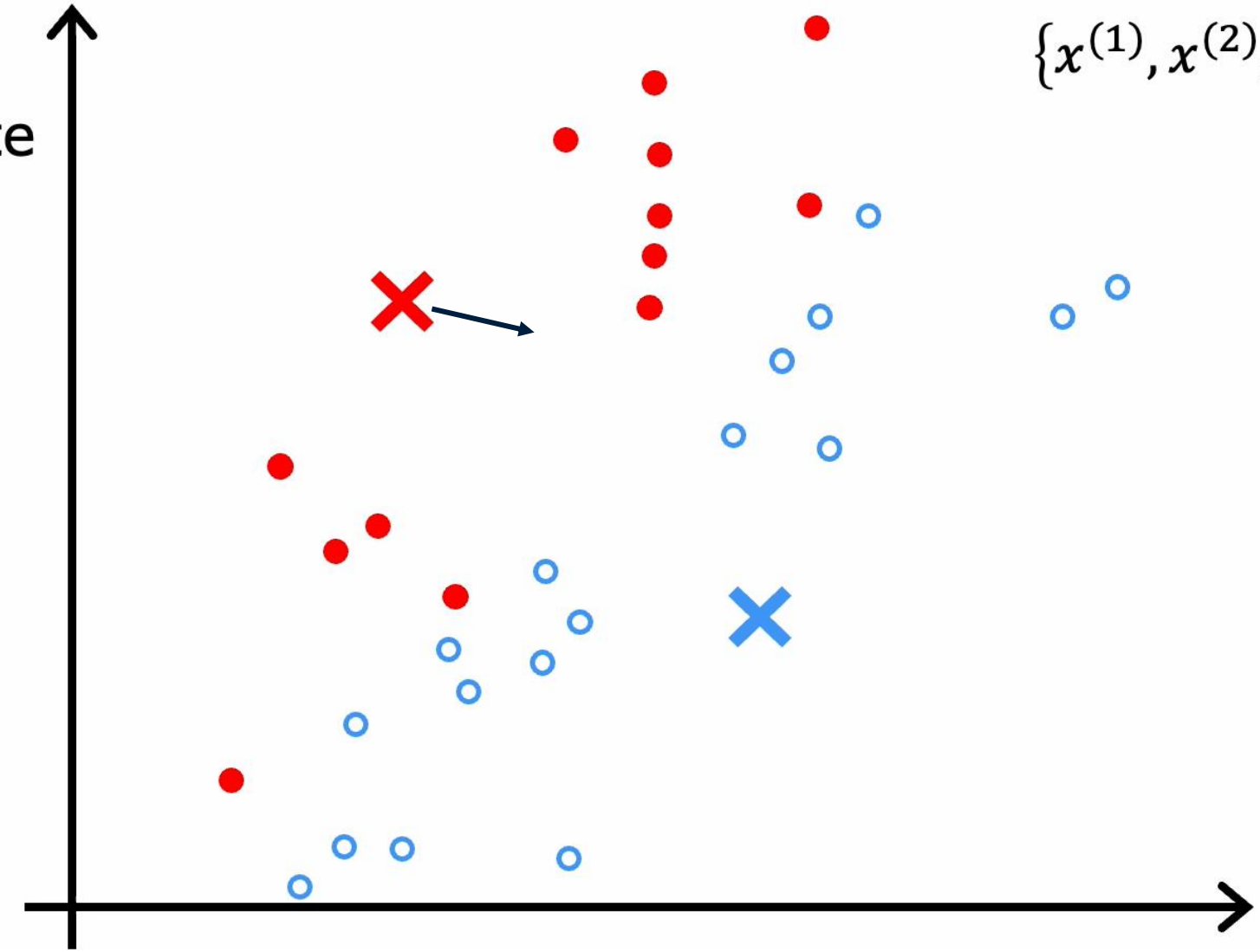

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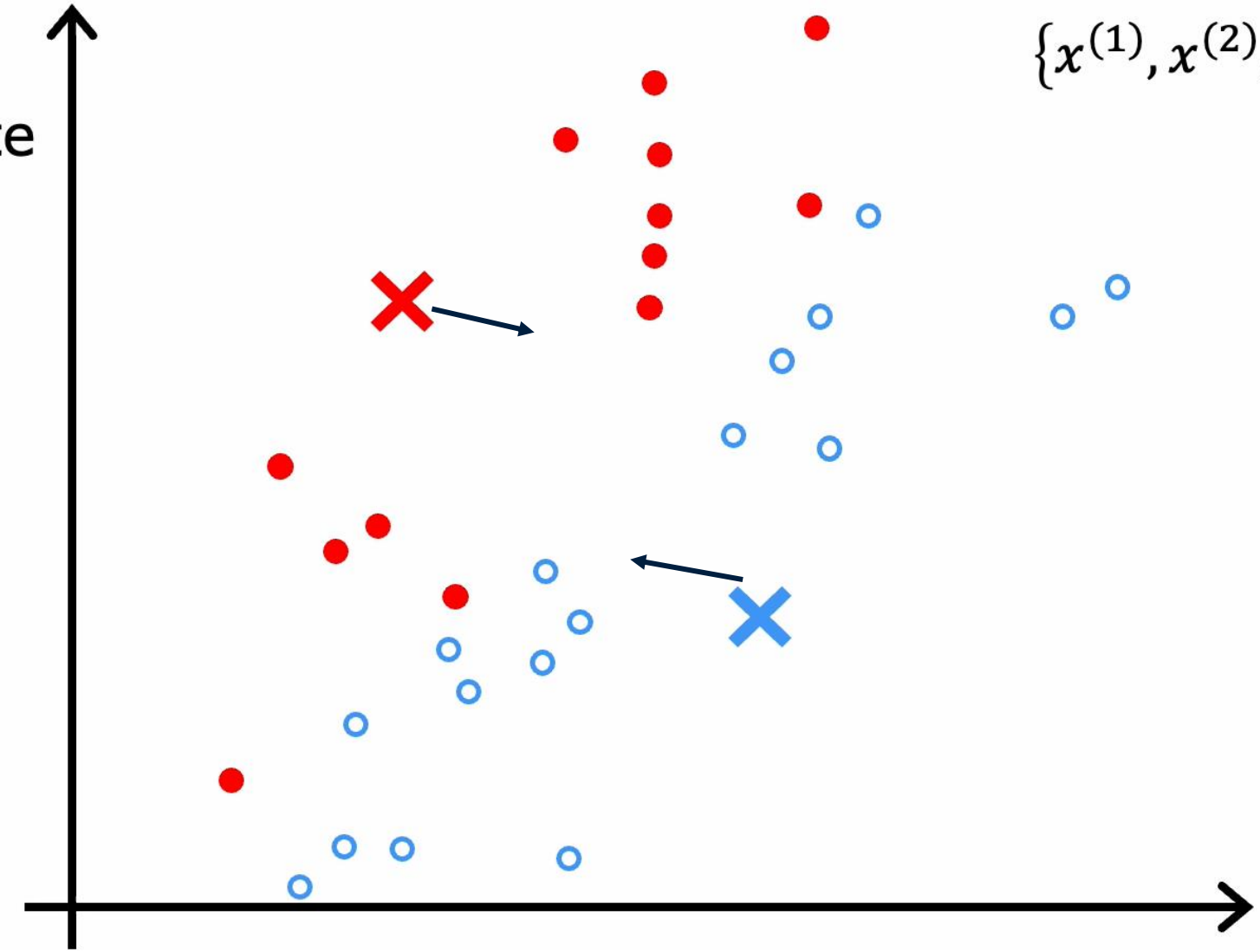

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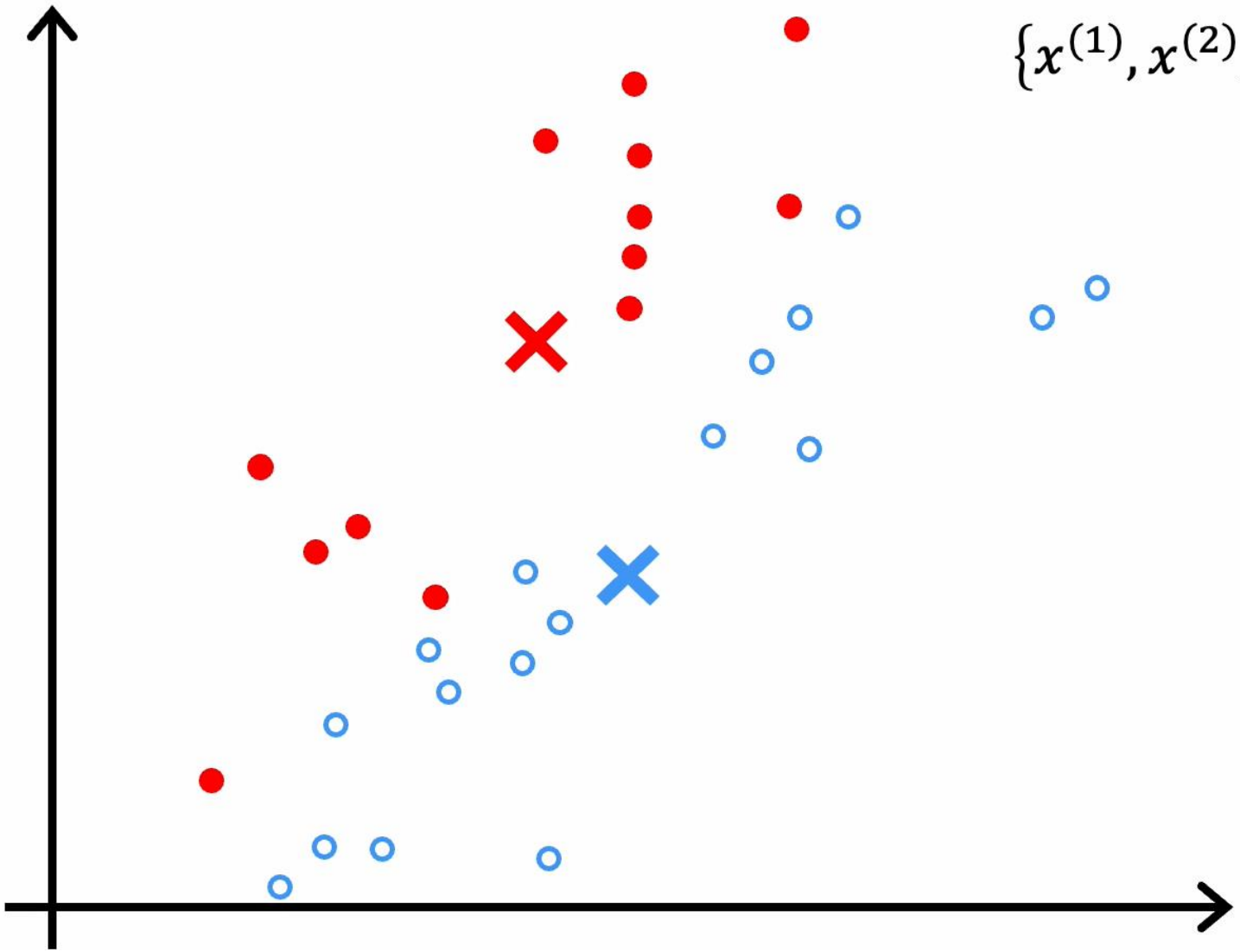

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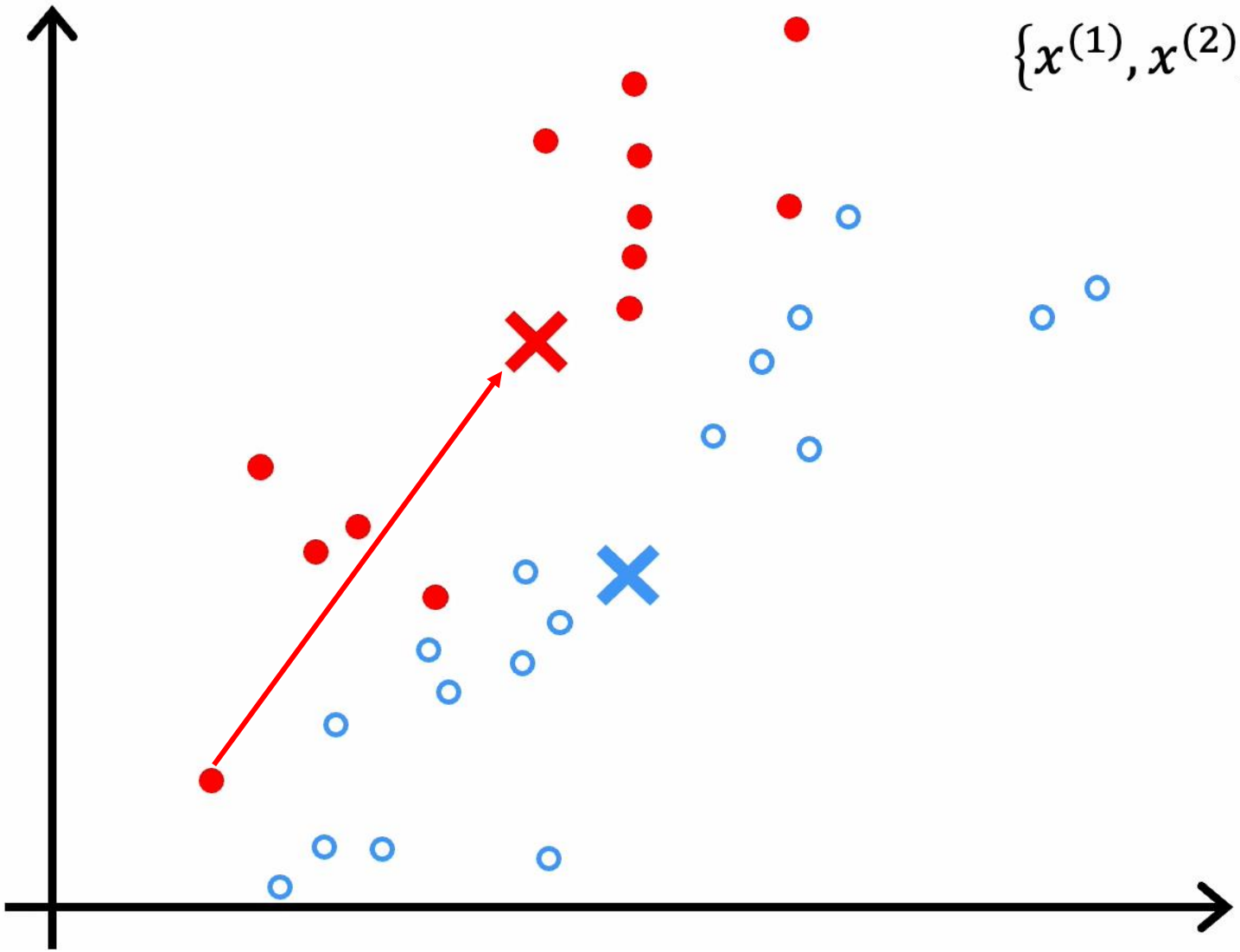

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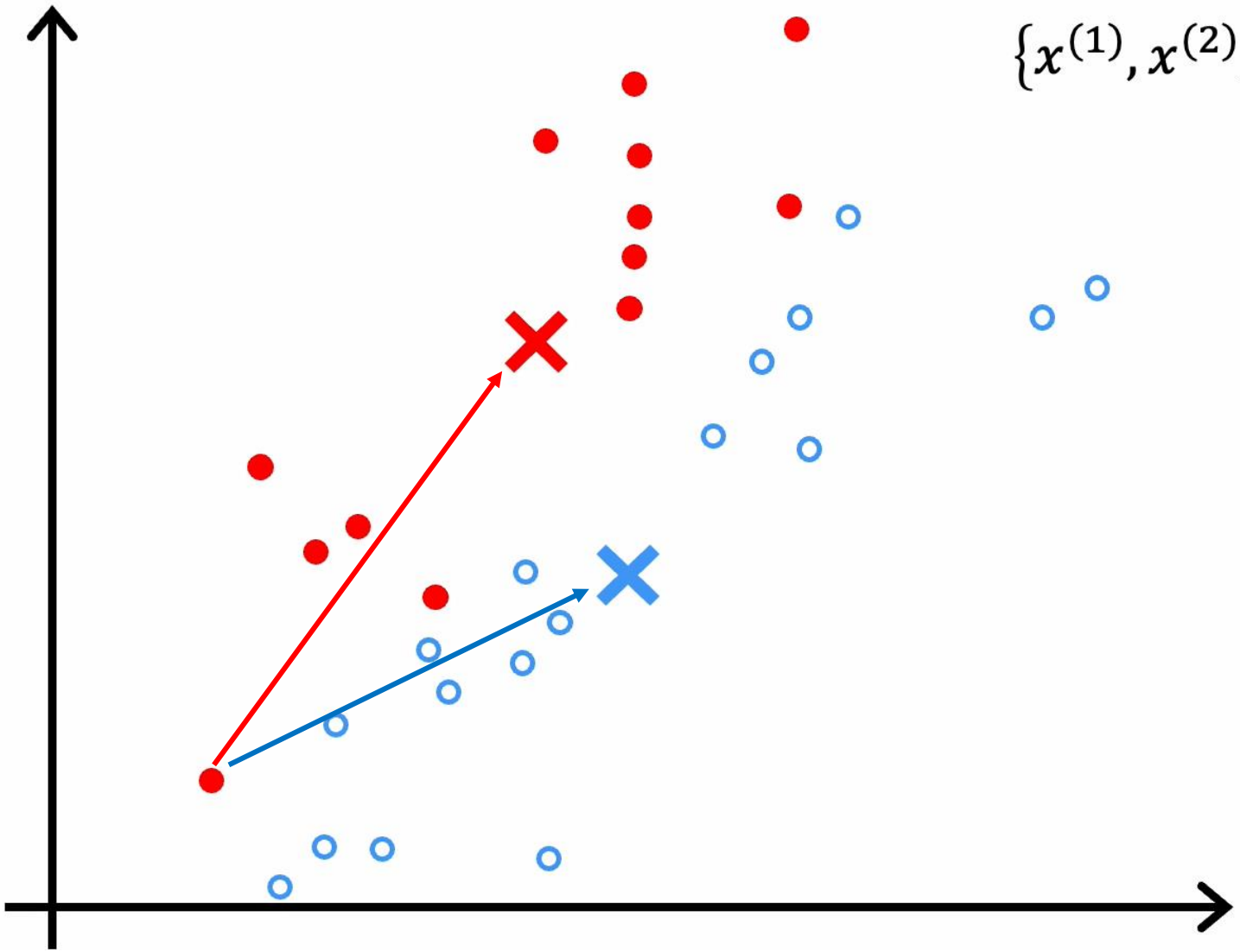

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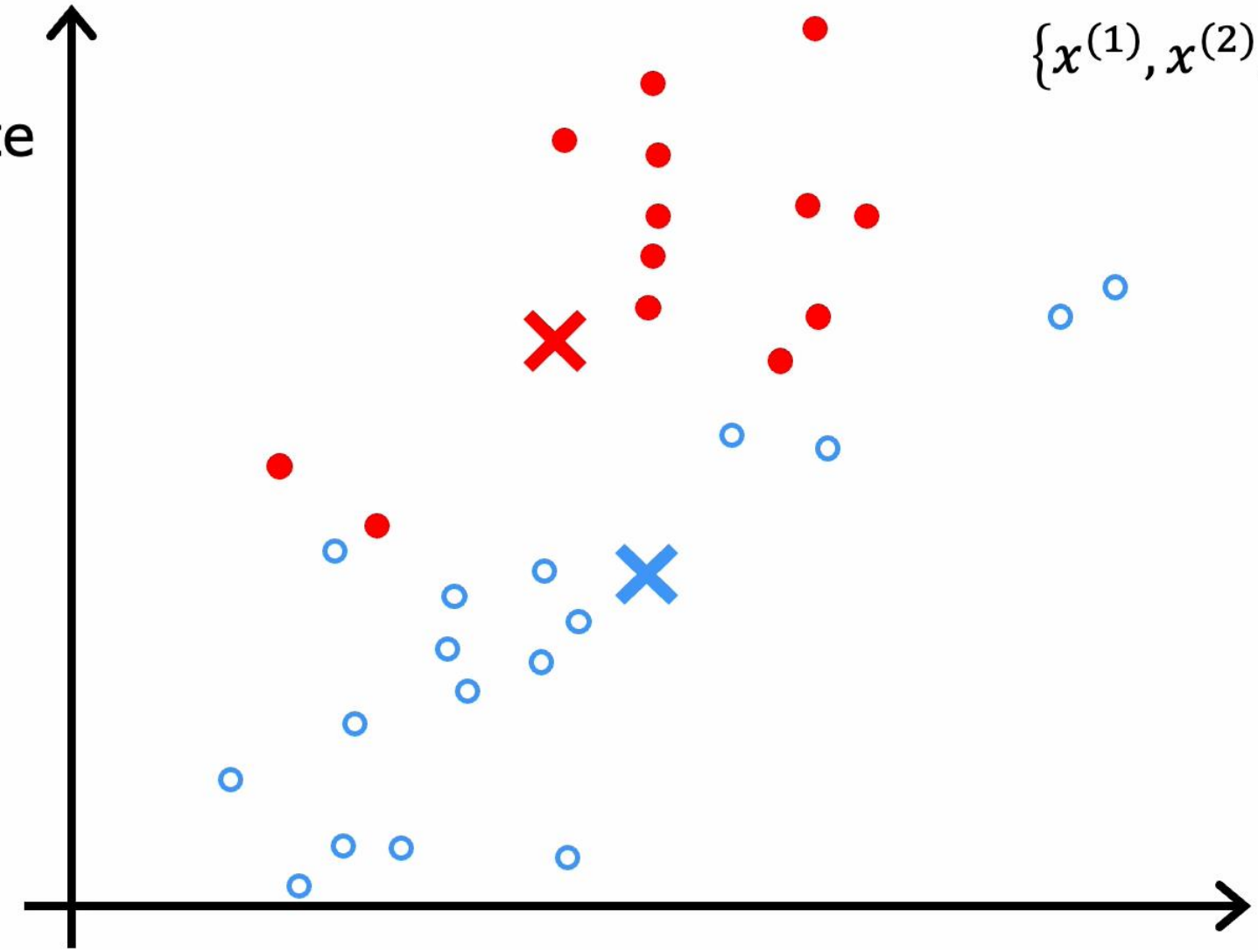

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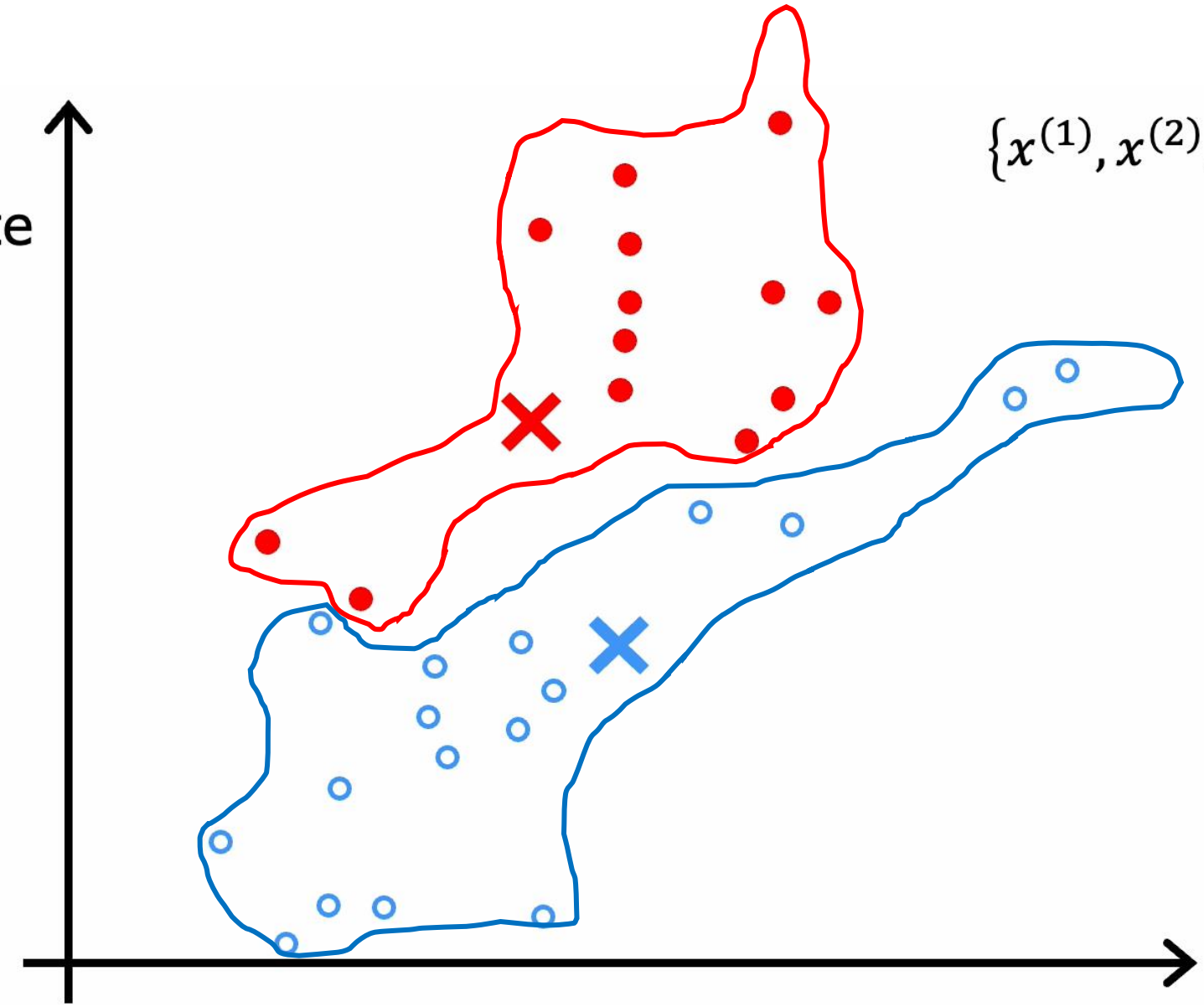

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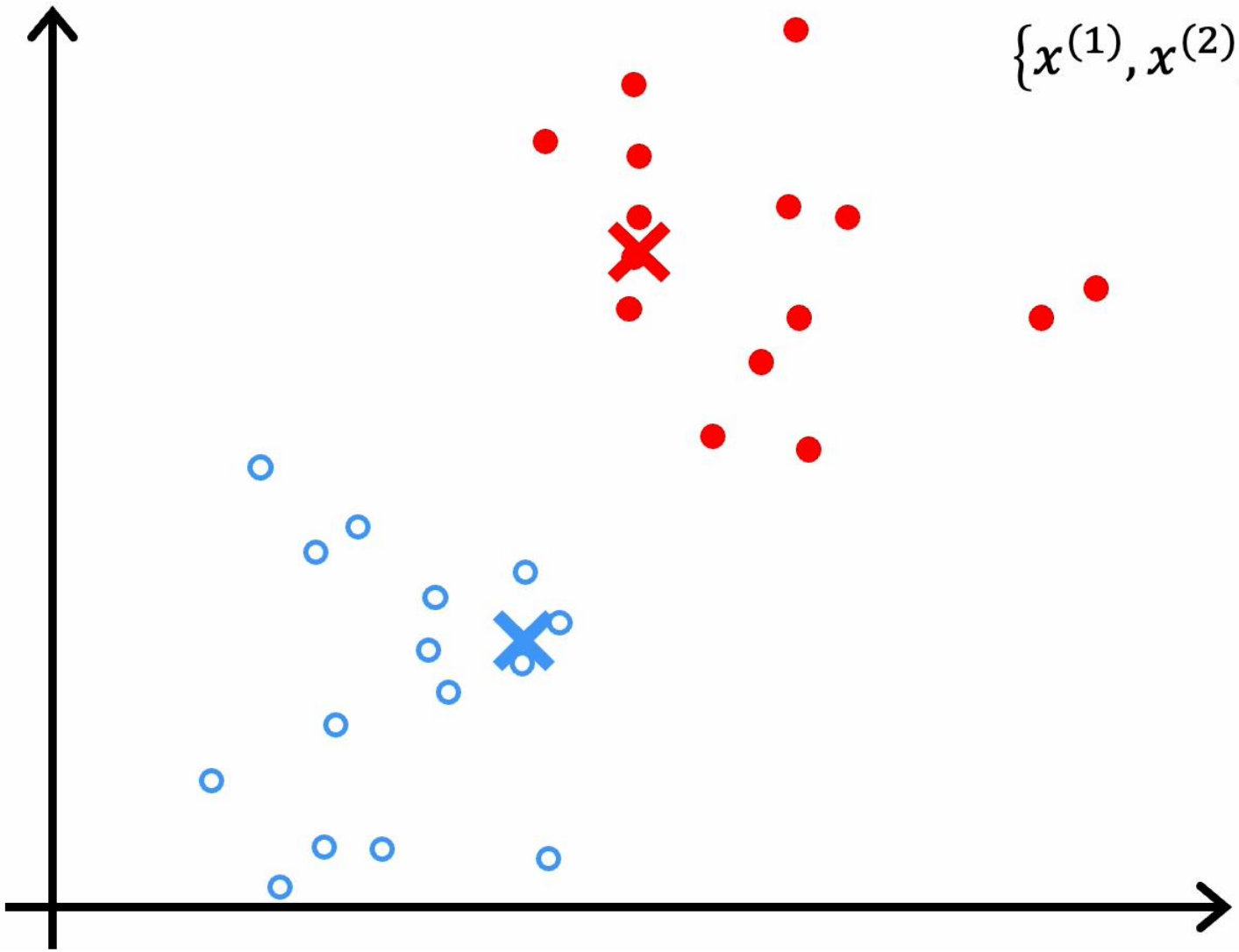

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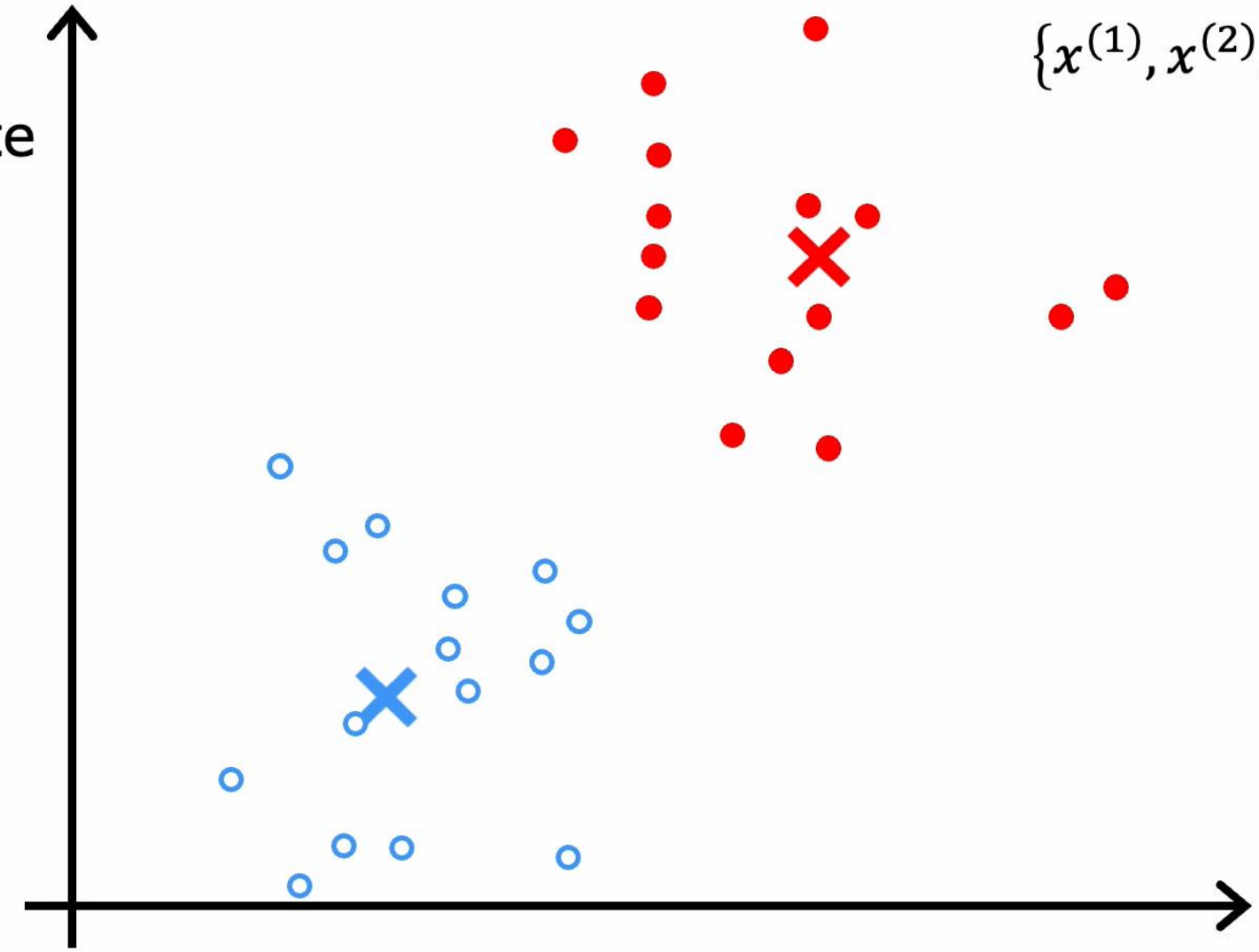

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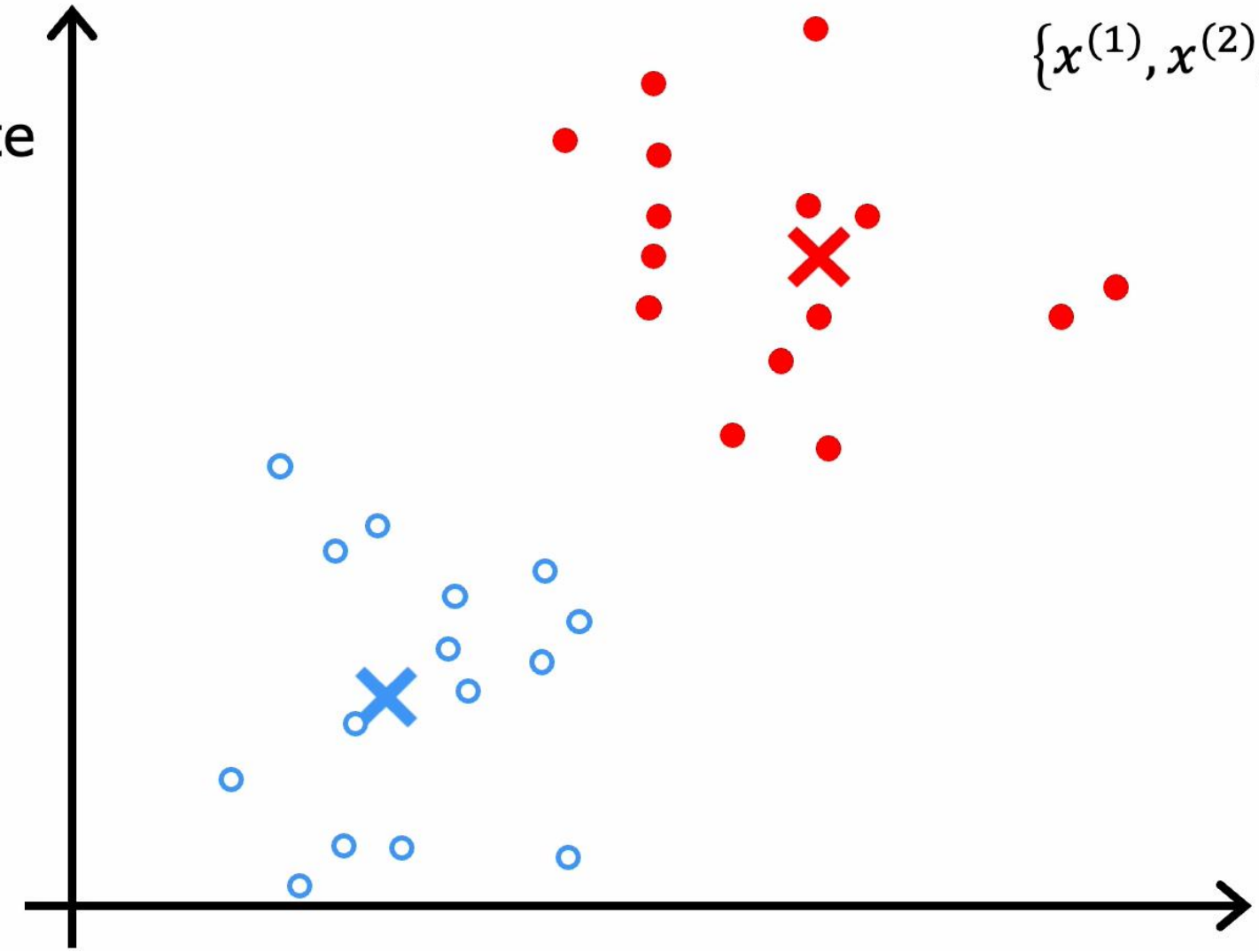

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Converged

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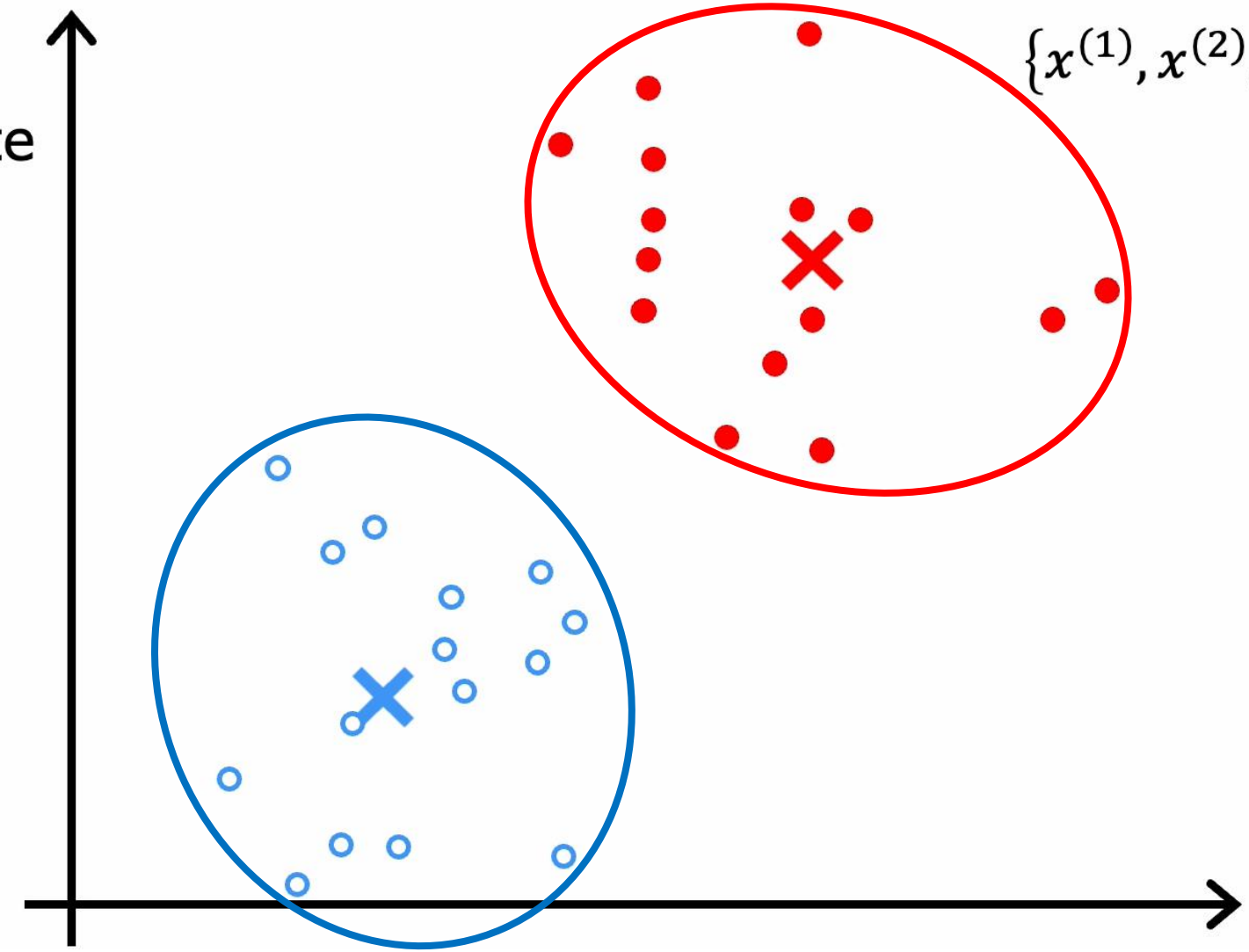

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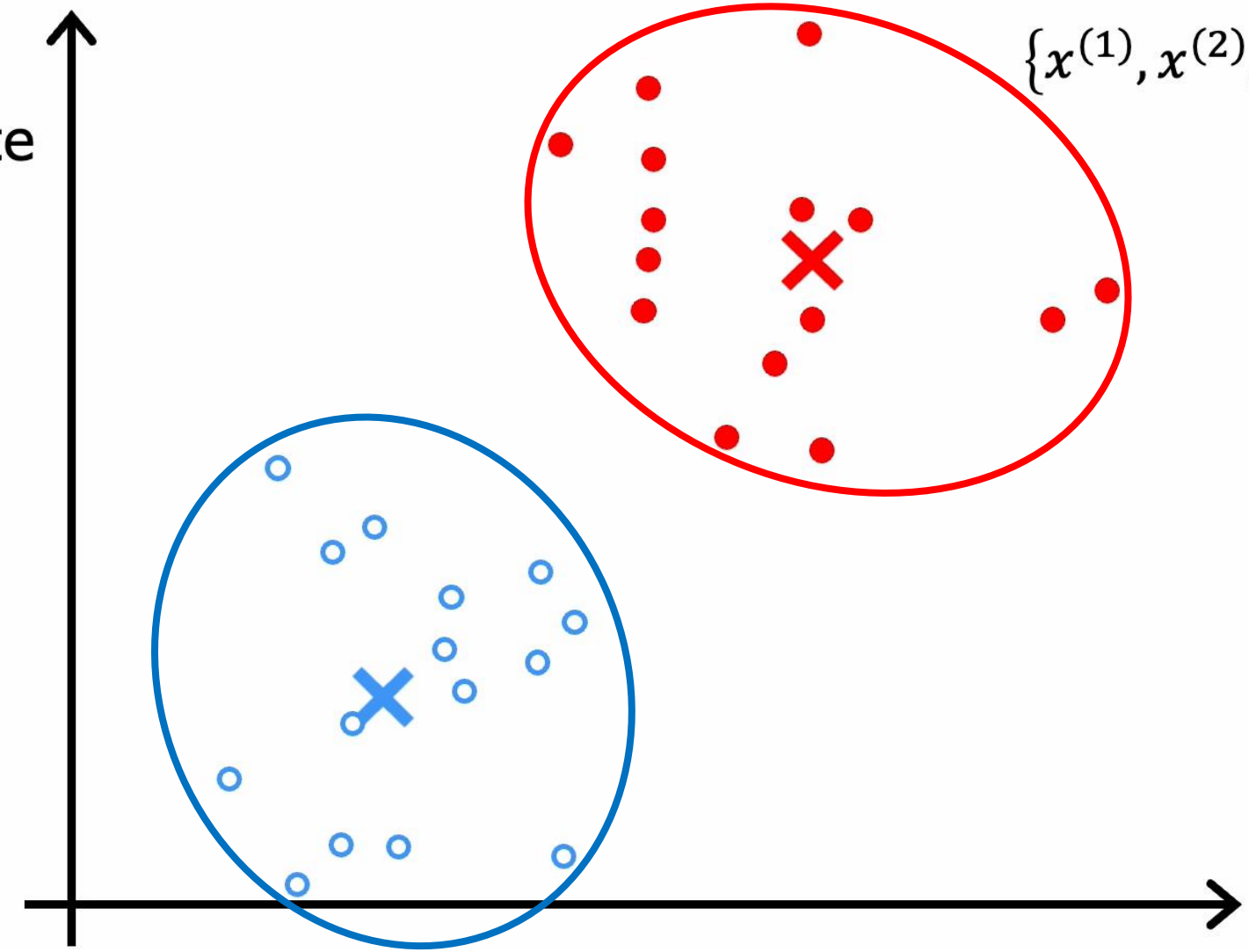

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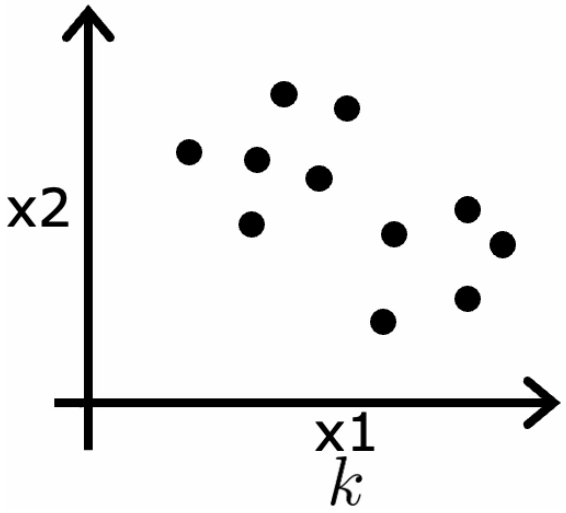
K-Means Algorithm



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K-Means Algorithm

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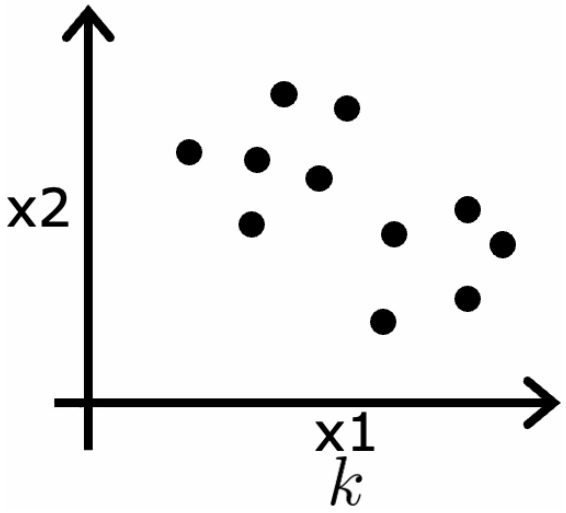

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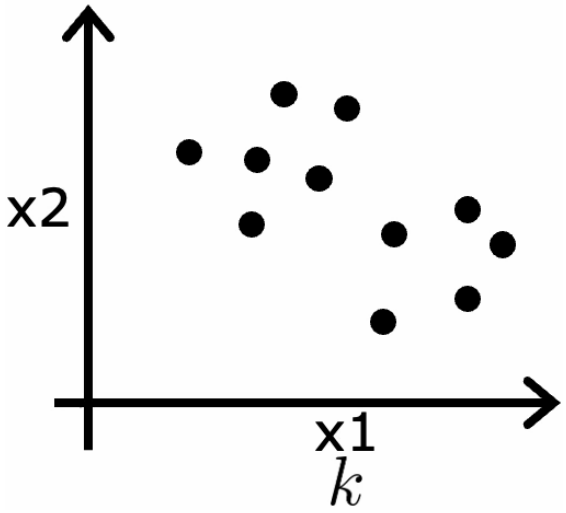

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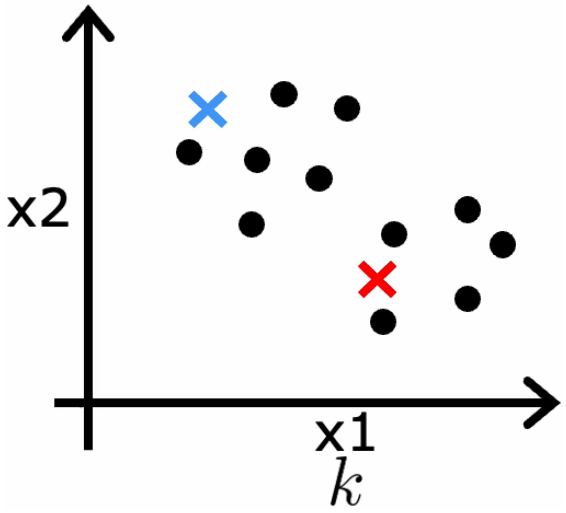

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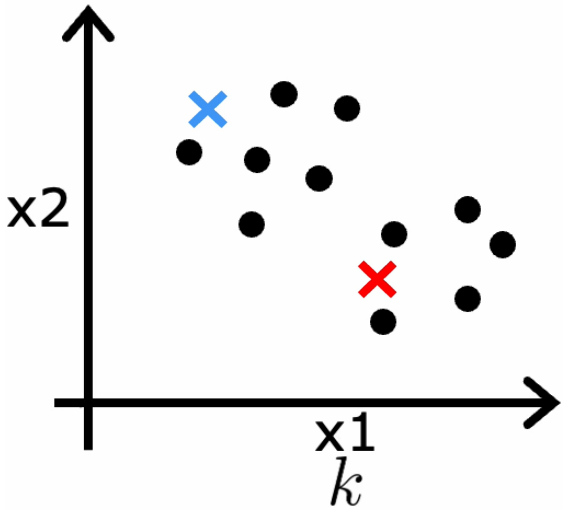
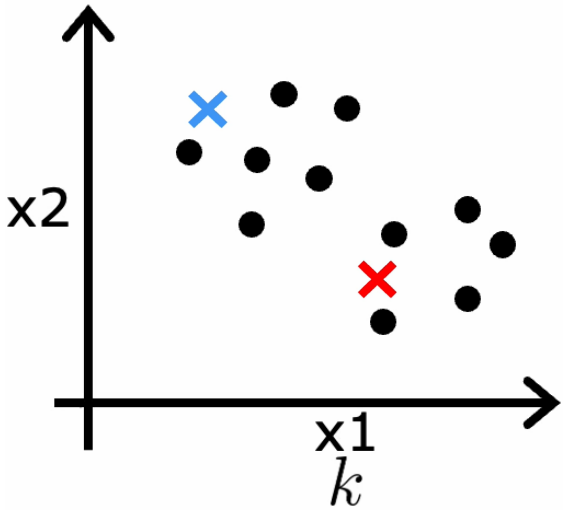


K-Means Algorithm

***K* = 2**

2-dimensional numbers *(x,y)*

Randomly initialize K cluster centroids $\boxed{\mu_1, \mu_2}, \ldots, \mu_K$



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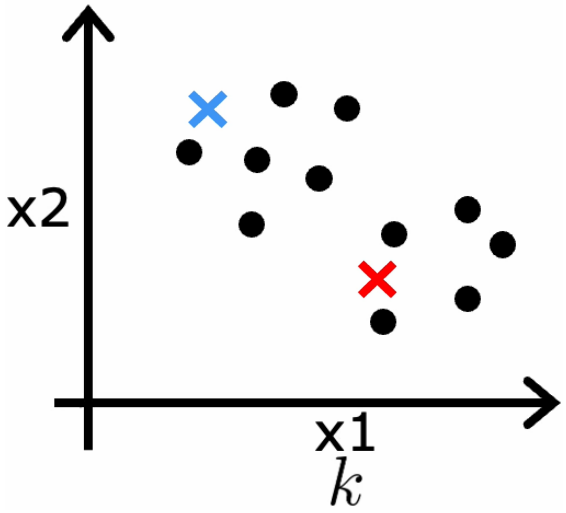
Repeat {

 # Assign points to cluster centroids



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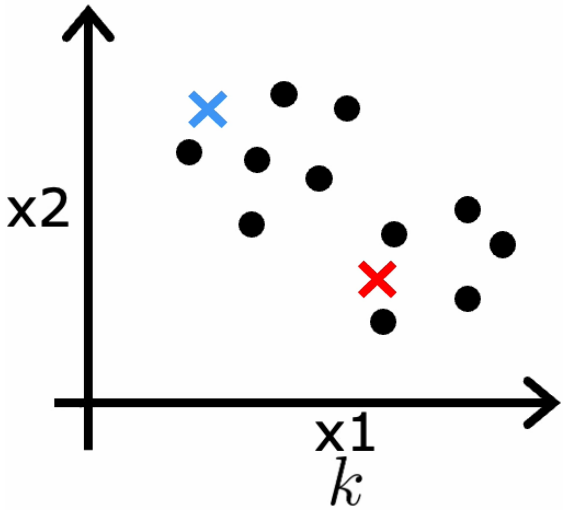

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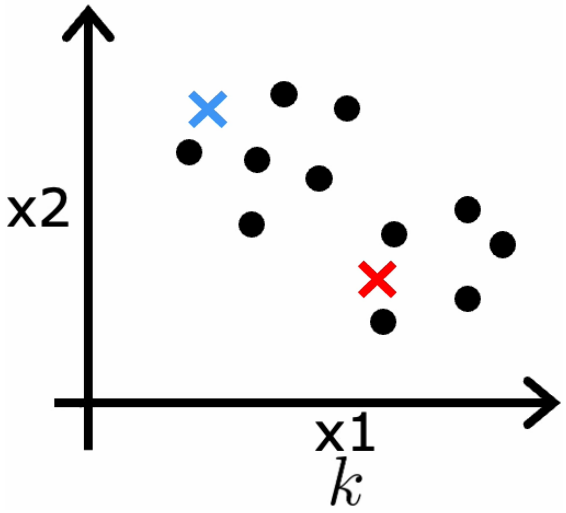

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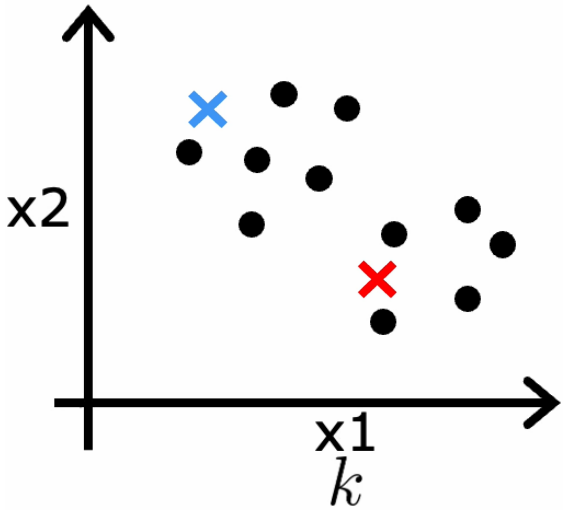

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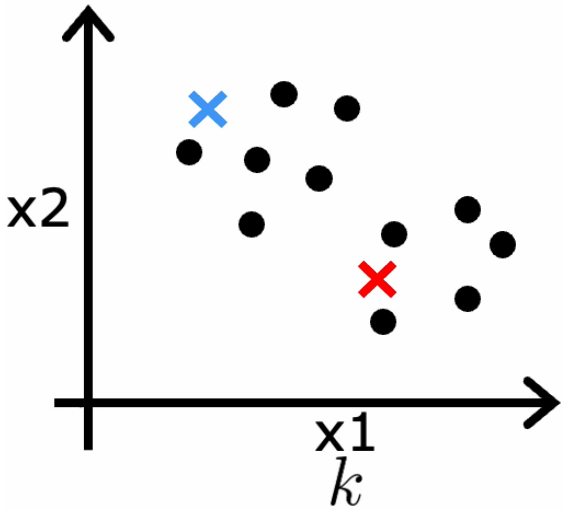

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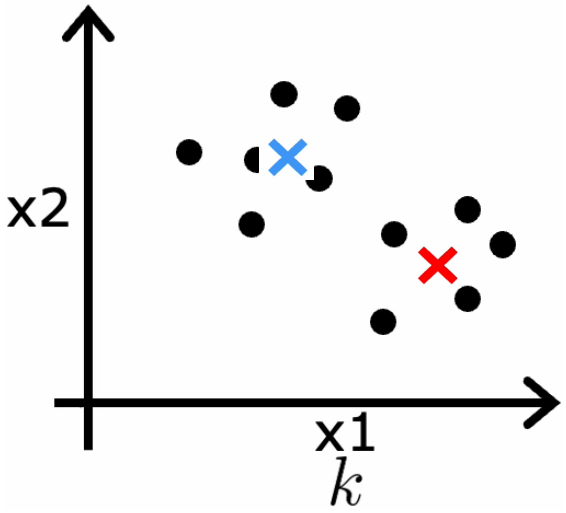
$$\mu_1 = \frac{1}{4}\left[x^{(1)} + x^{(5)} + x^{(6)} + x^{(10)}\right]$$



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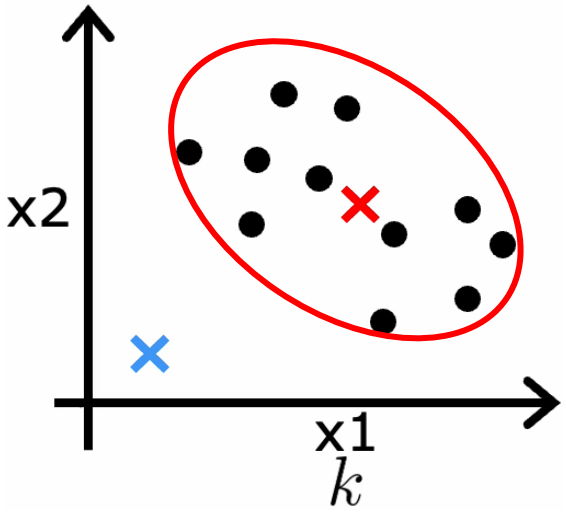

What if a cluster has no points assigned?

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K-Means Algorithm

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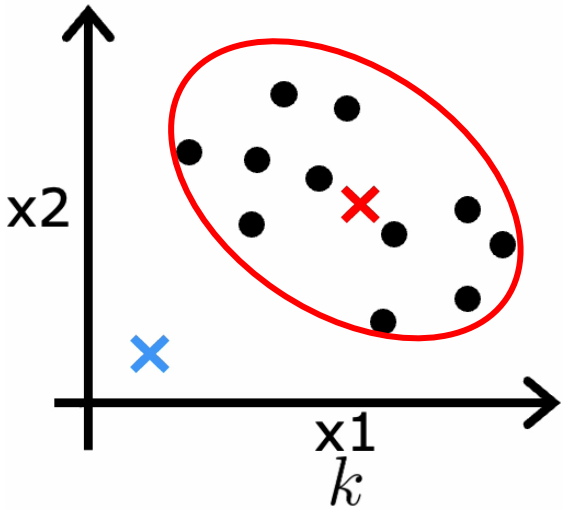

What if a cluster has no points assigned?
Either reinitialize, or
Eliminate and set *K = K - 1*

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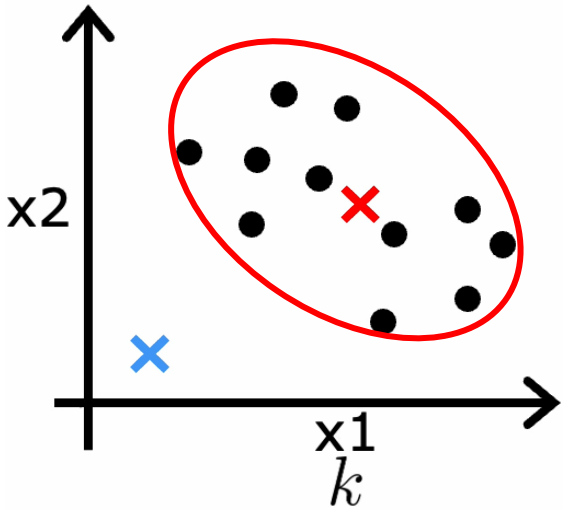

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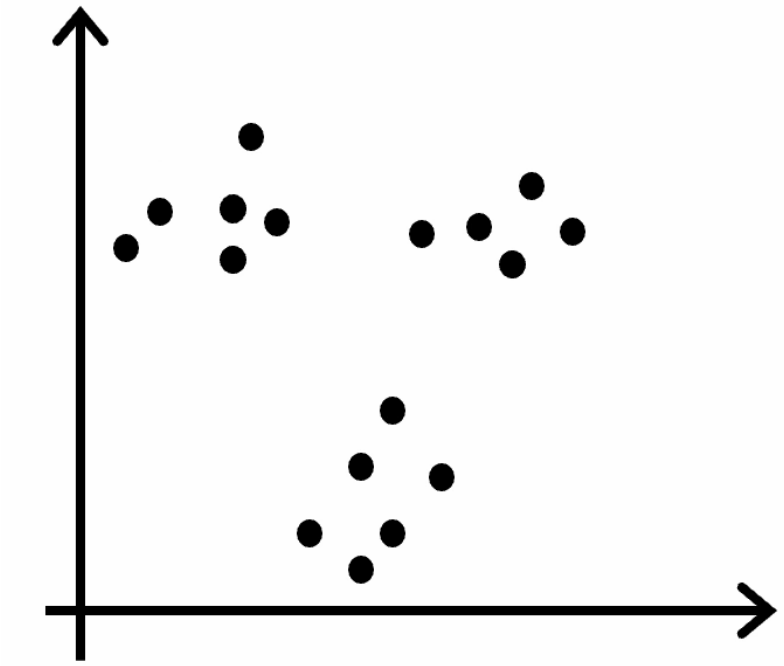
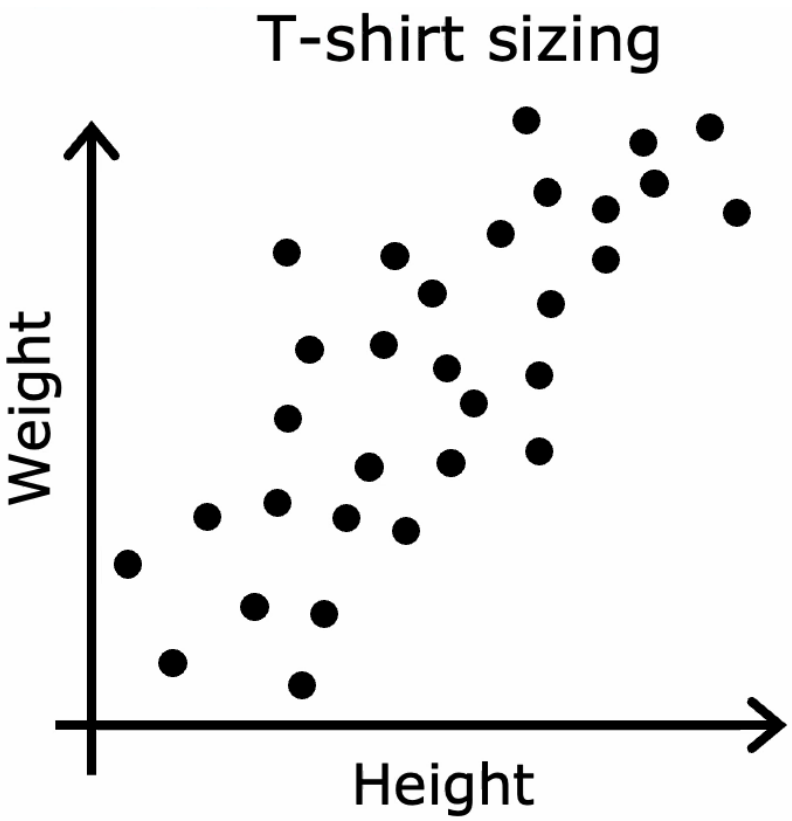
When Clusters not Well Separated?





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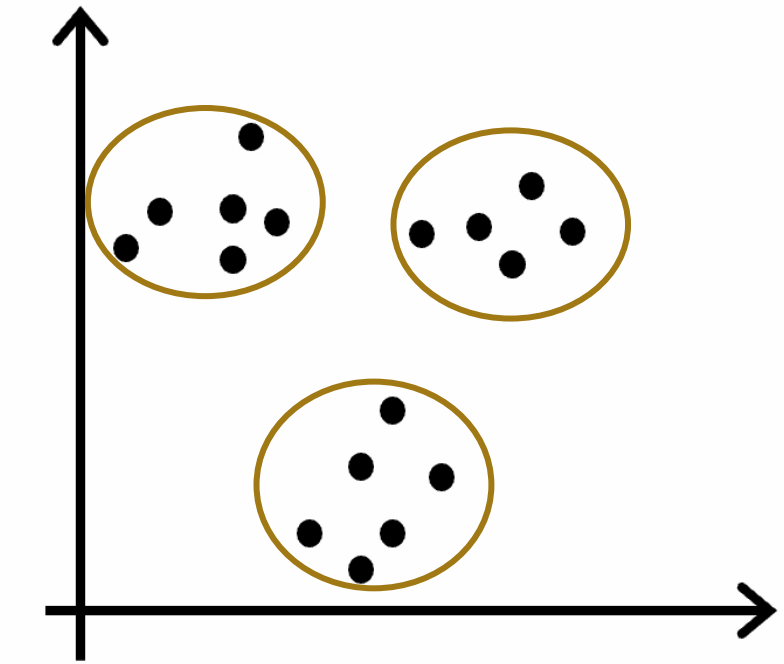
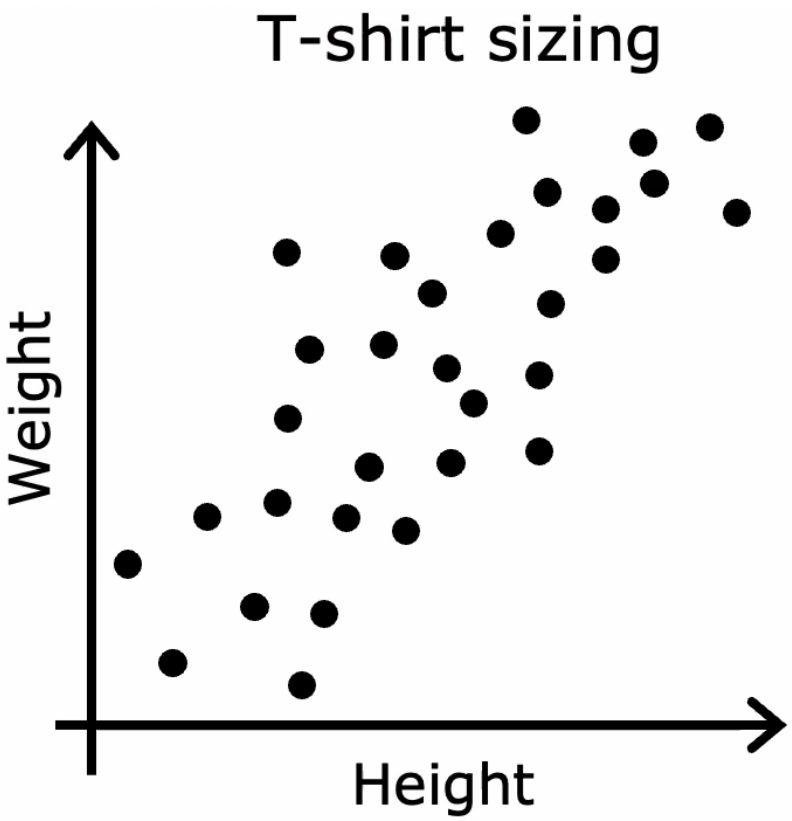
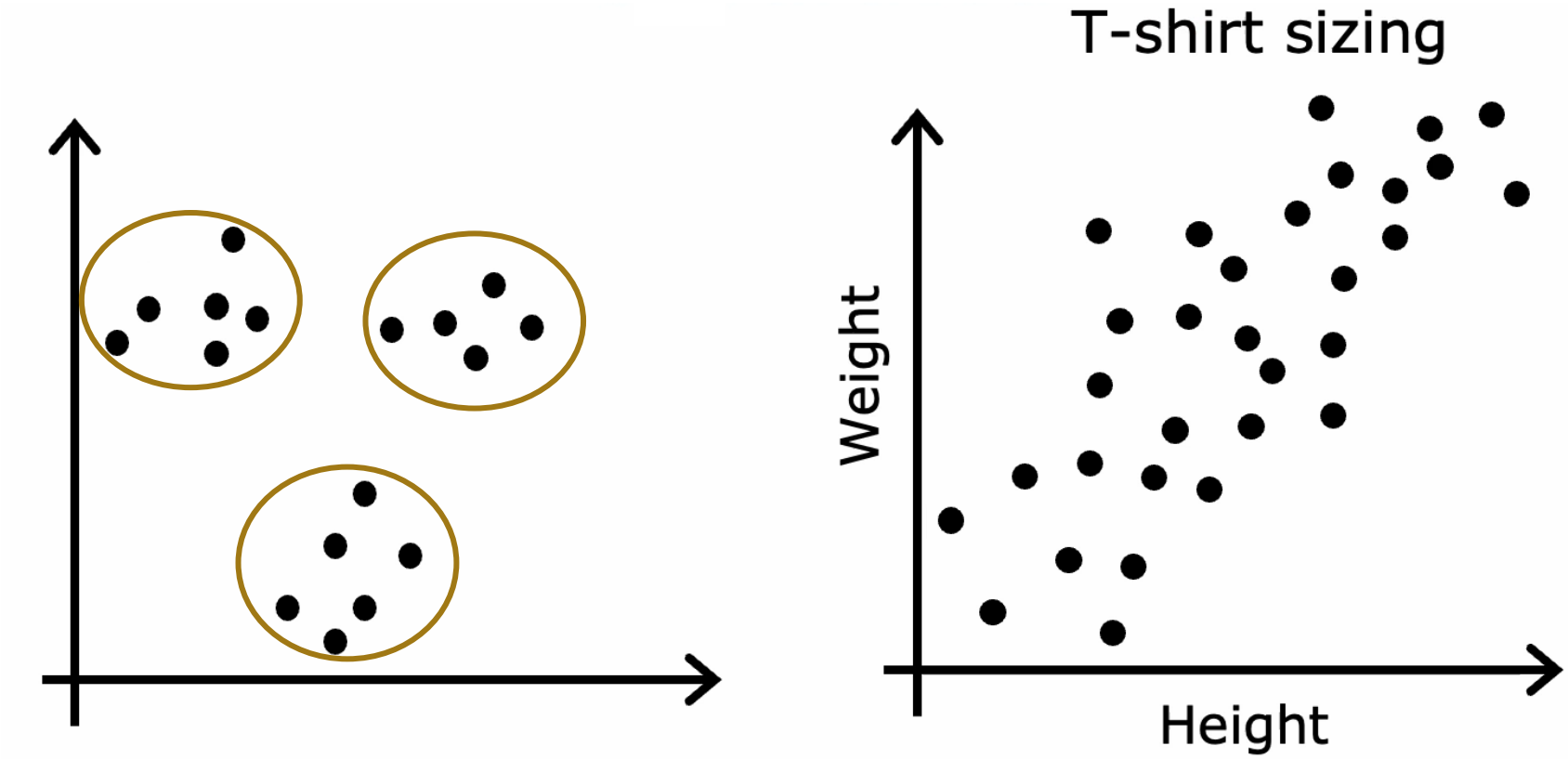




When Clusters not Well Separated?

S, M, L



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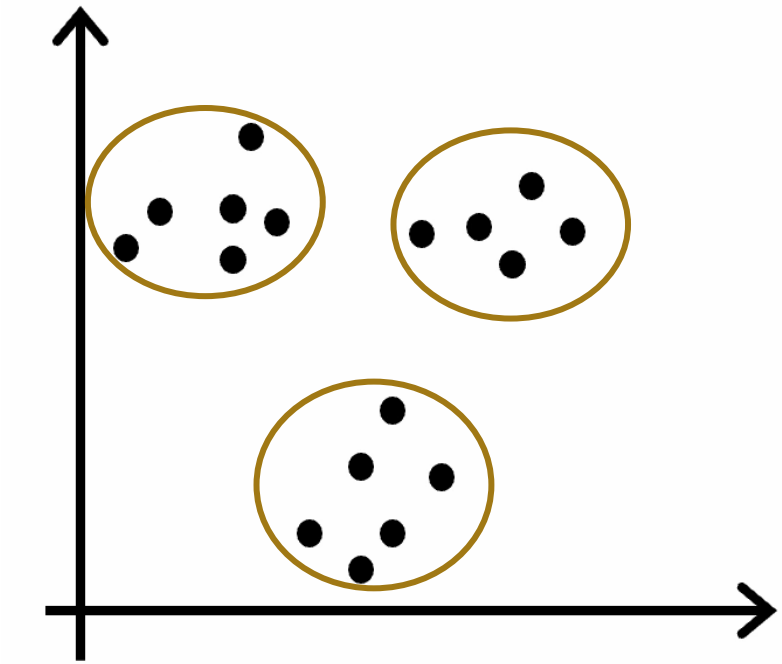

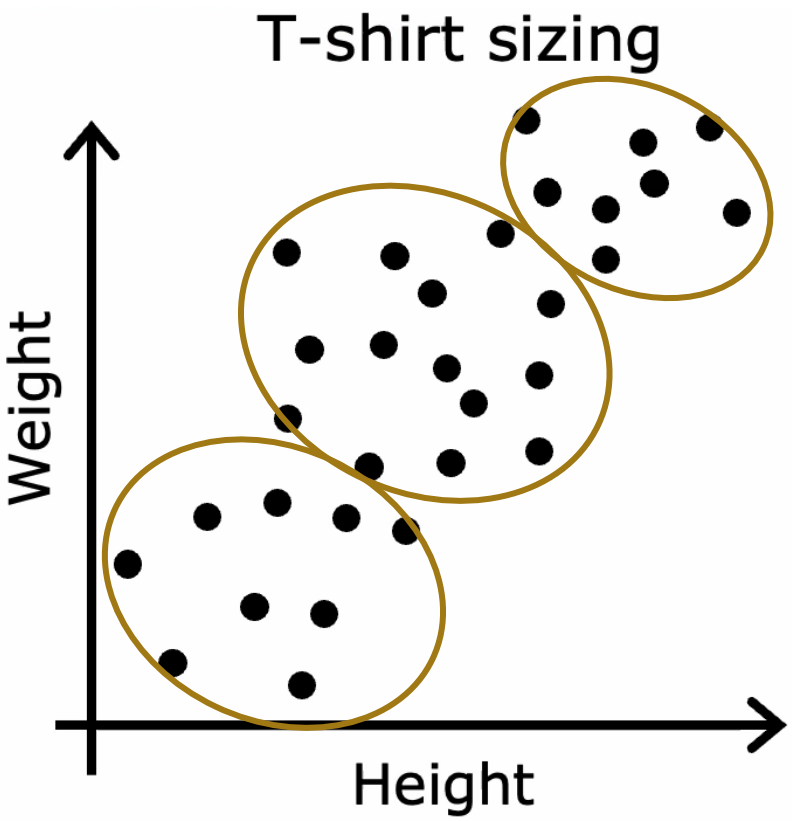

K-Means Optimization Objective

$c^{(i)}$ = index of cluster $(1, 2, \ldots, K)$ to which example $x^{(i)}$ is currently assigned

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μ_k = cluster centroid k

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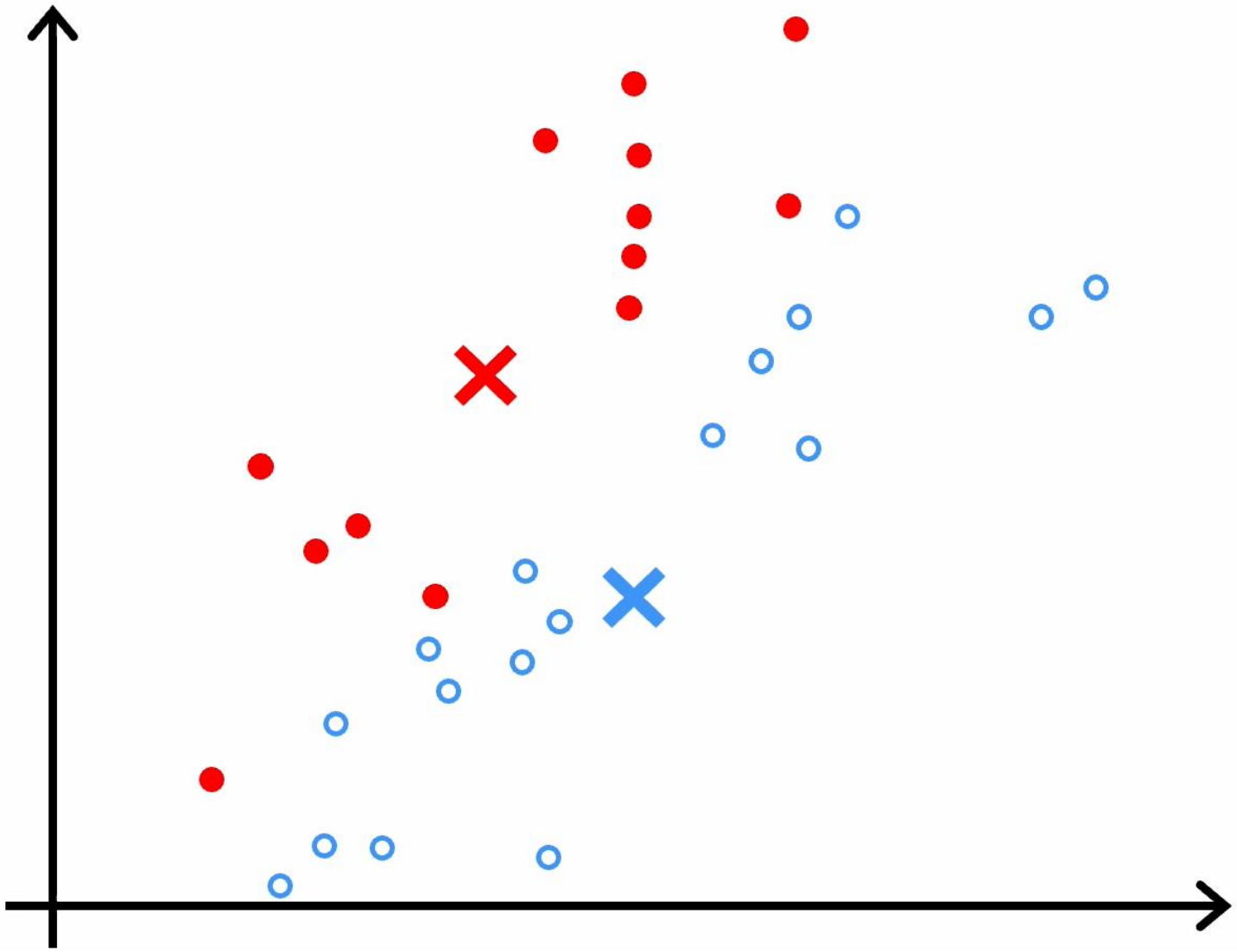
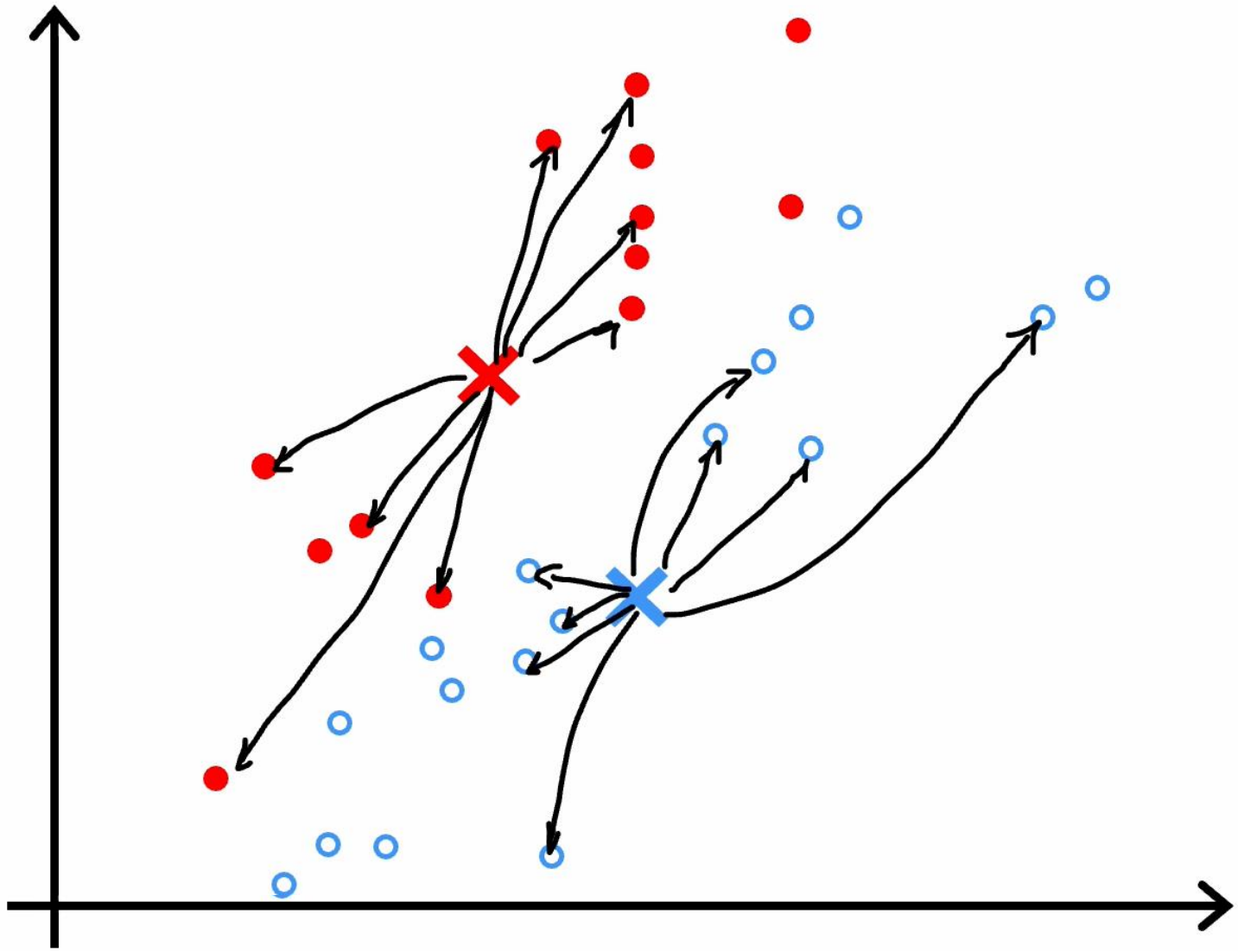
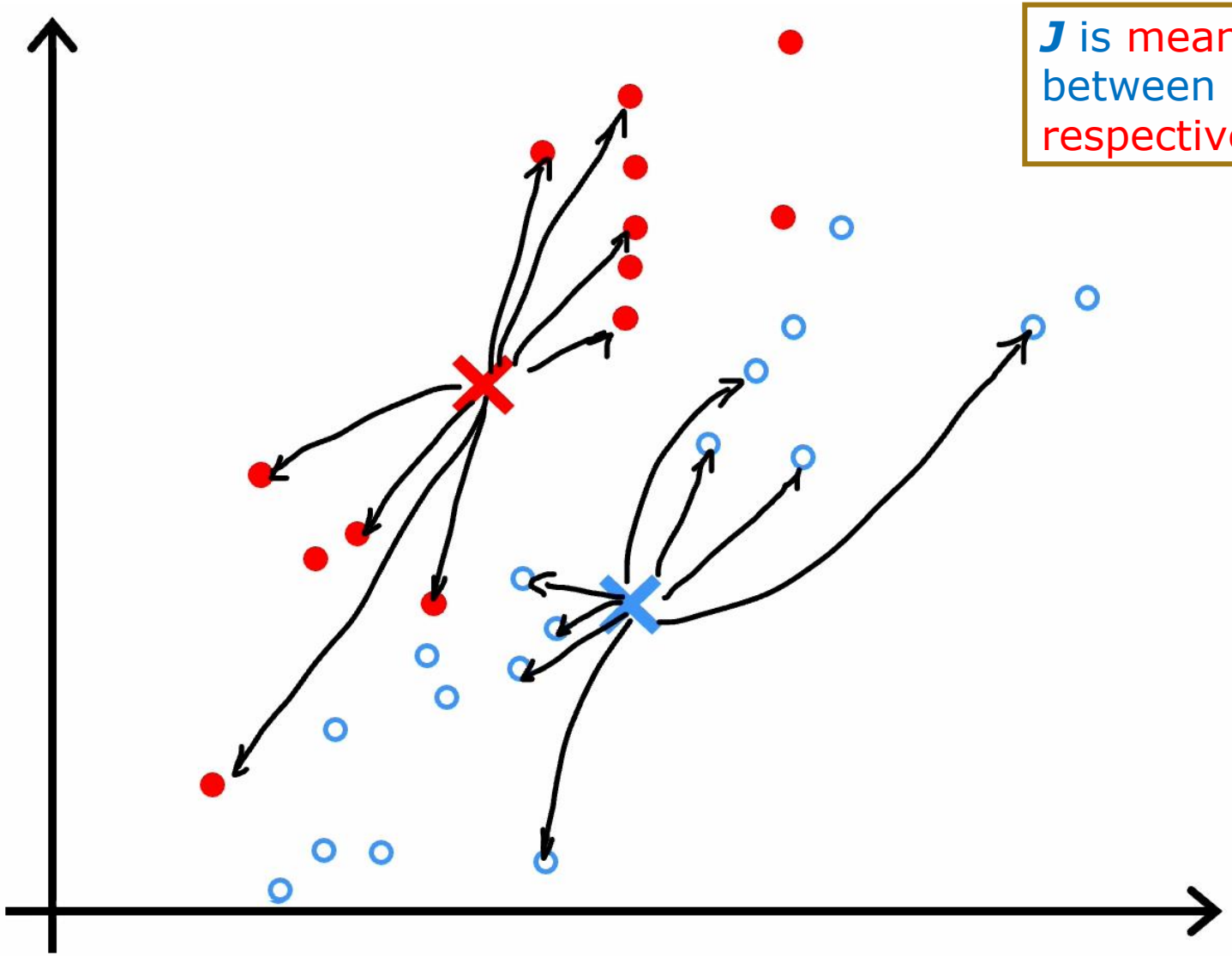
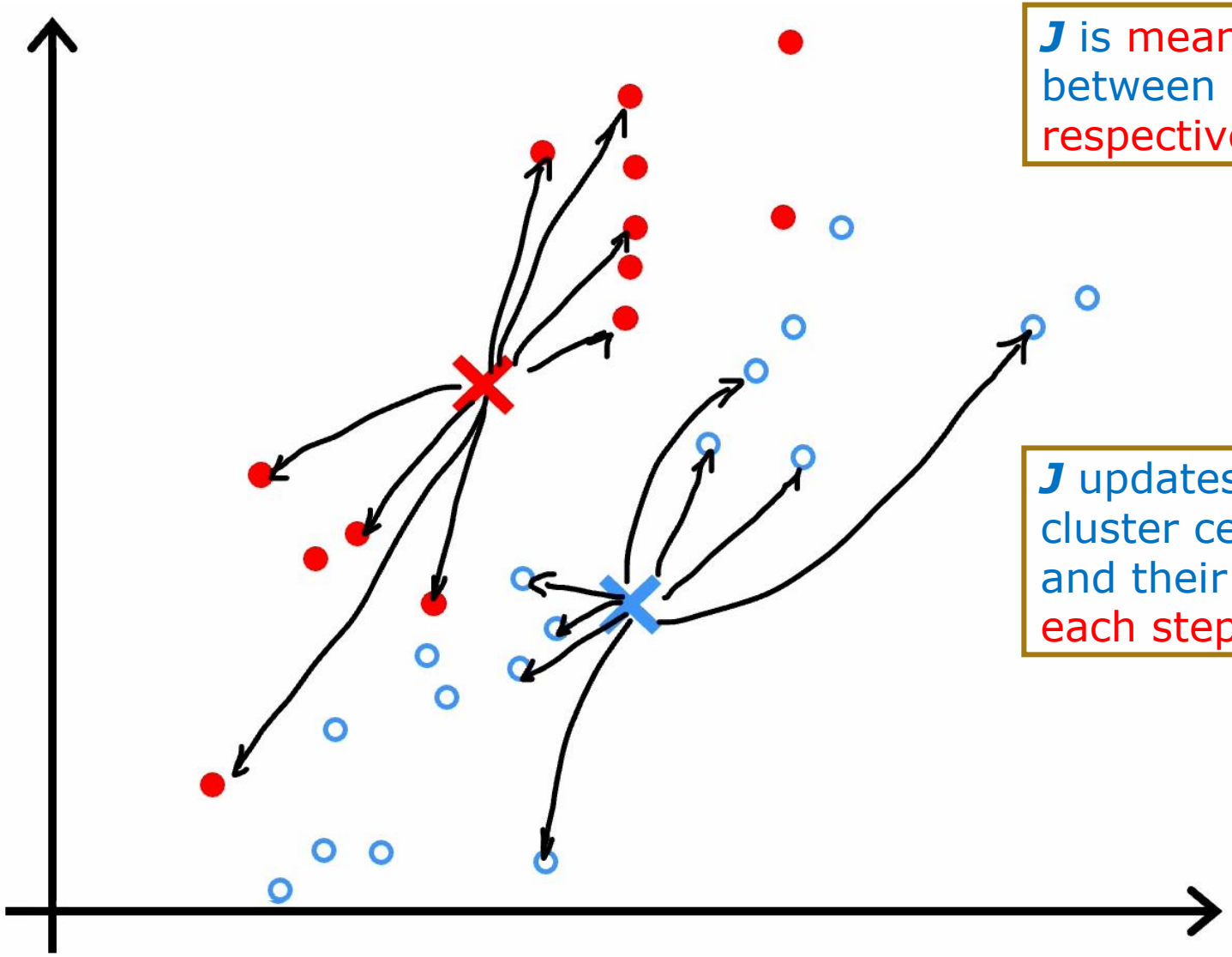
Cost function





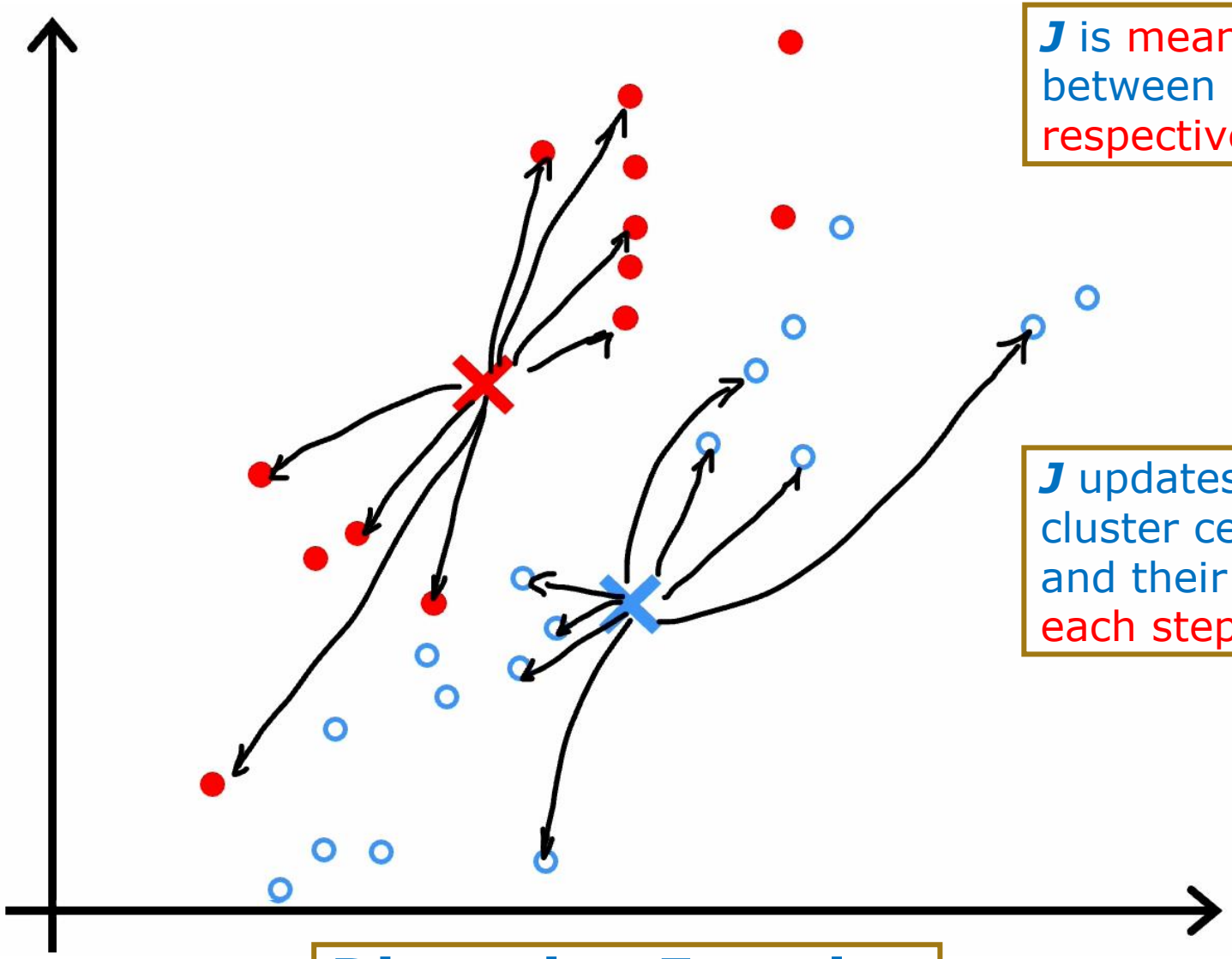


J is mean of squared distances between all points and their respective cluster centroids



J is mean of squared distances between all points and their respective cluster centroids

J updates the assignment of cluster centroids $c^{(1)}$, $c^{(2)}$,… and their locations μ_1, μ_2,… at each step to reduce itself.



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Distortion Function

K-Means Optimization Objective

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μ_k = cluster centroid k

$\mu_{c^{(i)}}$ = cluster centroid of cluster to which example $x^{(i)}$ has been assigned

Cost function

$$J(c^{(1)}, \ldots, c^{(m)}, \mu_1, \ldots, \mu_K) = \frac{1}{m}\sum_{i=1}^{m}\left\|x^{(i)} - \mu_{c^{(i)}}\right\|^2$$

$$\min_{\substack{c^{(1)}, \ldots, c^{(m)} \\ \mu_1, \ldots, \mu_K}} J(c^{(1)}, \ldots, c^{(m)}, \mu_1, \ldots, \mu_K)$$

Cost Function for K-Means

$$J\left(c^{(1)}, \ldots, c^{(m)}, \mu_1, \ldots, \mu_K\right) = \frac{1}{m}\sum_{i=1}^{m}\left\|x^{(i)} - \mu_{c^{(i)}}\right\|^2$$

Repeat {

 # Assign points to cluster centroids

for i = 1 to m

$c^{(i)}$:= index of cluster centroid closest to $x^{(i)}$

 # Move cluster centroids

for k = 1 to K

μ_k := average of points in cluster k

}

Cost Function for K-Means

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}

Part-I

Cost Function for K-Means

$$J(\underbrace{c^{(1)}, \ldots, c^{(m)}}_{\text{Update}}, \mu_1, \ldots, \mu_K) = \frac{1}{m}\sum_{i=1}^{m}\left\|x^{(i)} - \mu_{c^{(i)}}\right\|^2$$

Repeat {

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Cost Function for K-Means

$$J(\underbrace{c^{(1)}, \ldots, c^{(m)}}_{\text{Update}}, \underbrace{\mu_1, \ldots, \mu_K}_{\text{Fixed}}) = \frac{1}{m}\sum_{i=1}^{m}\left\|x^{(i)} - \mu_{c^{(i)}}\right\|^2$$

Repeat {

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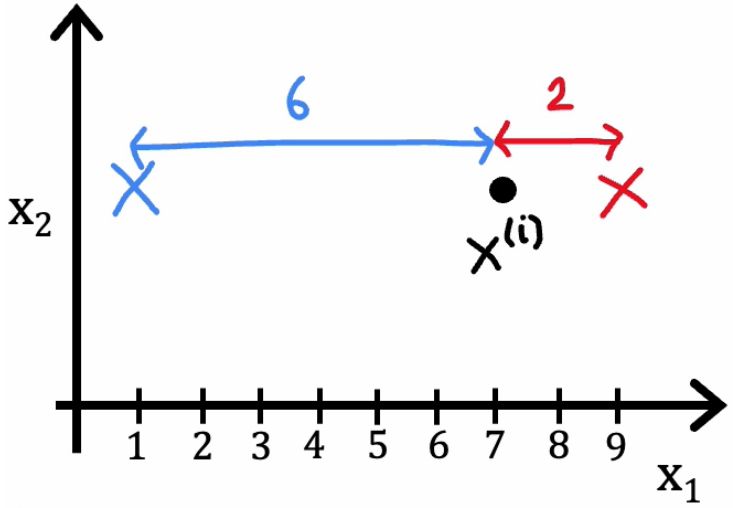

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Part-II

Cost Function for K-Means

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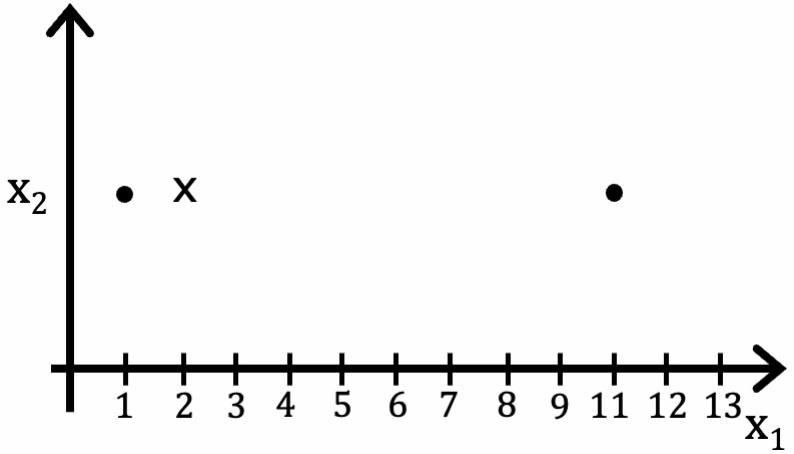
Part-II



Cost Function for K-Means

$\frac{1}{2}(1^2 + 9^2) = \frac{1}{2}(1+81) = 41$

$$J(\underbrace{c^{(1)}, \ldots, c^{(m)}}_{\text{Fixed}}, \underbrace{\mu_1, \ldots, \mu_K}_{\text{Update}}) = \frac{1}{m}\sum_{i=1}^{m}\left\|x^{(i)} - \mu_{c^{(i)}}\right\|^2$$

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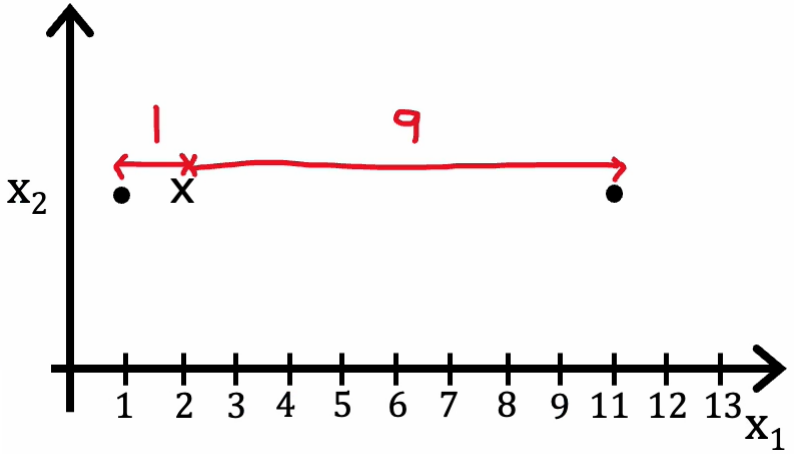

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Cost Function for K-Means

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$\frac{1}{2}(1^2 + 9^2) = \frac{1}{2}(1+81) = 41$

$\frac{1}{2}(1 + 11) = 6$

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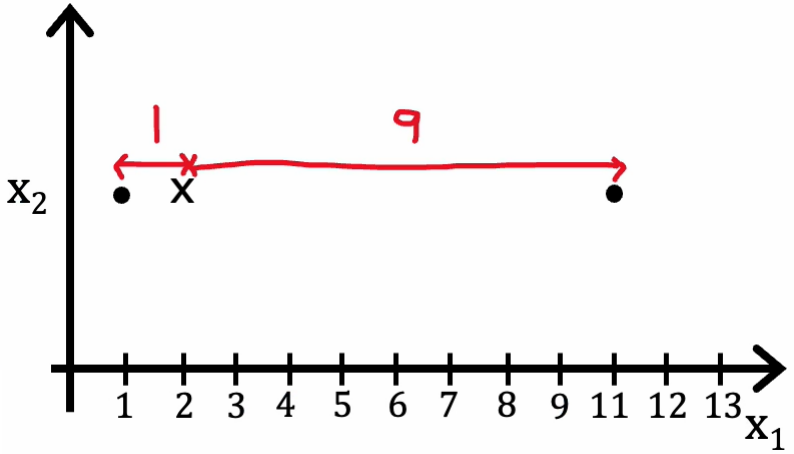

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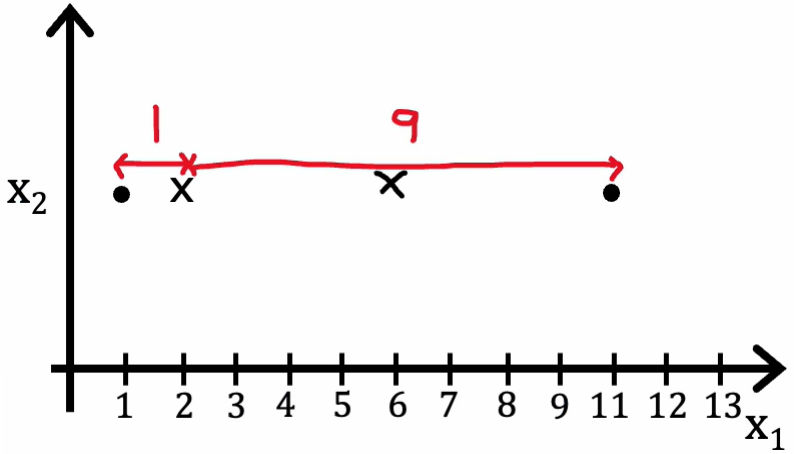

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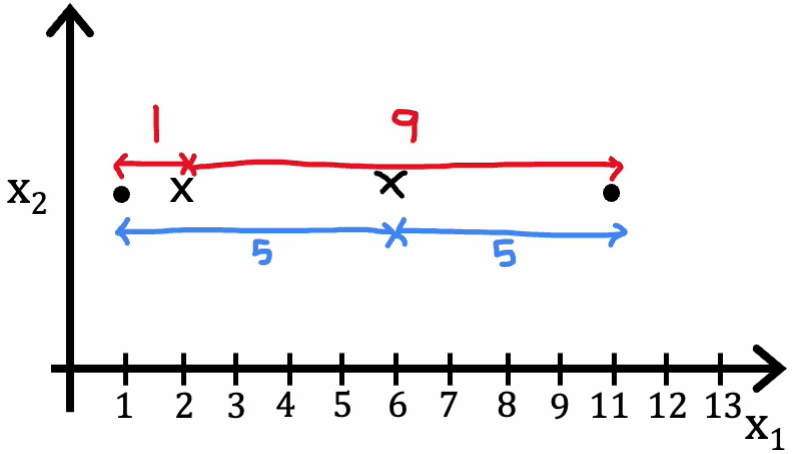
}

Part-II



Cost Function for K-Means

$$J(\underbrace{c^{(1)}, \ldots, c^{(m)}}_{\text{Fixed}}, \underbrace{\mu_1, \ldots, \mu_K}_{\text{Update}}) = \frac{1}{m}\sum_{i=1}^{m}\left\|x^{(i)} - \mu_{c^{(i)}}\right\|^2$$

Repeat {

 # Assign points to cluster centroids

for i = 1 to m

$c^{(i)}$:= index of cluster centroid closest to $x^{(i)}$

 # Move cluster centroids

for k = 1 to K

μ_k := average of points in cluster k

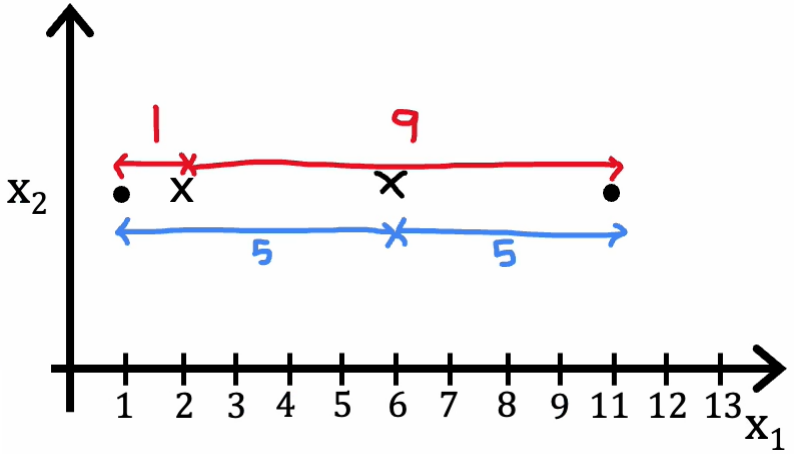
}

Part-II

$\frac{1}{2}(1^2 + 9^2) = \frac{1}{2}(1+81) = 41$

$\frac{1}{2}(1 + 11) = 6$

$\frac{1}{2}(5^2 + 5^2) = 25$



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Part-II

Convergence?

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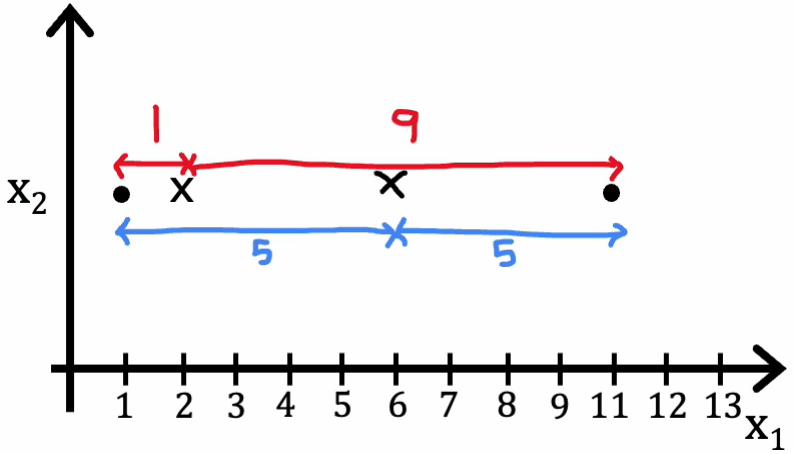

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Part-II

Convergence?

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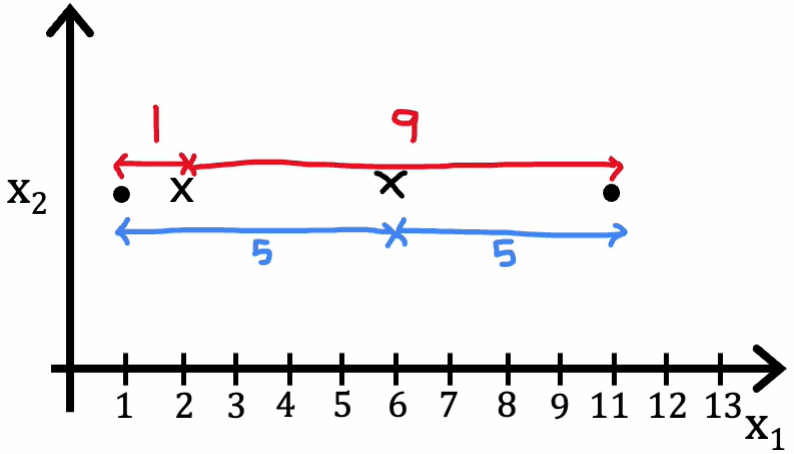

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